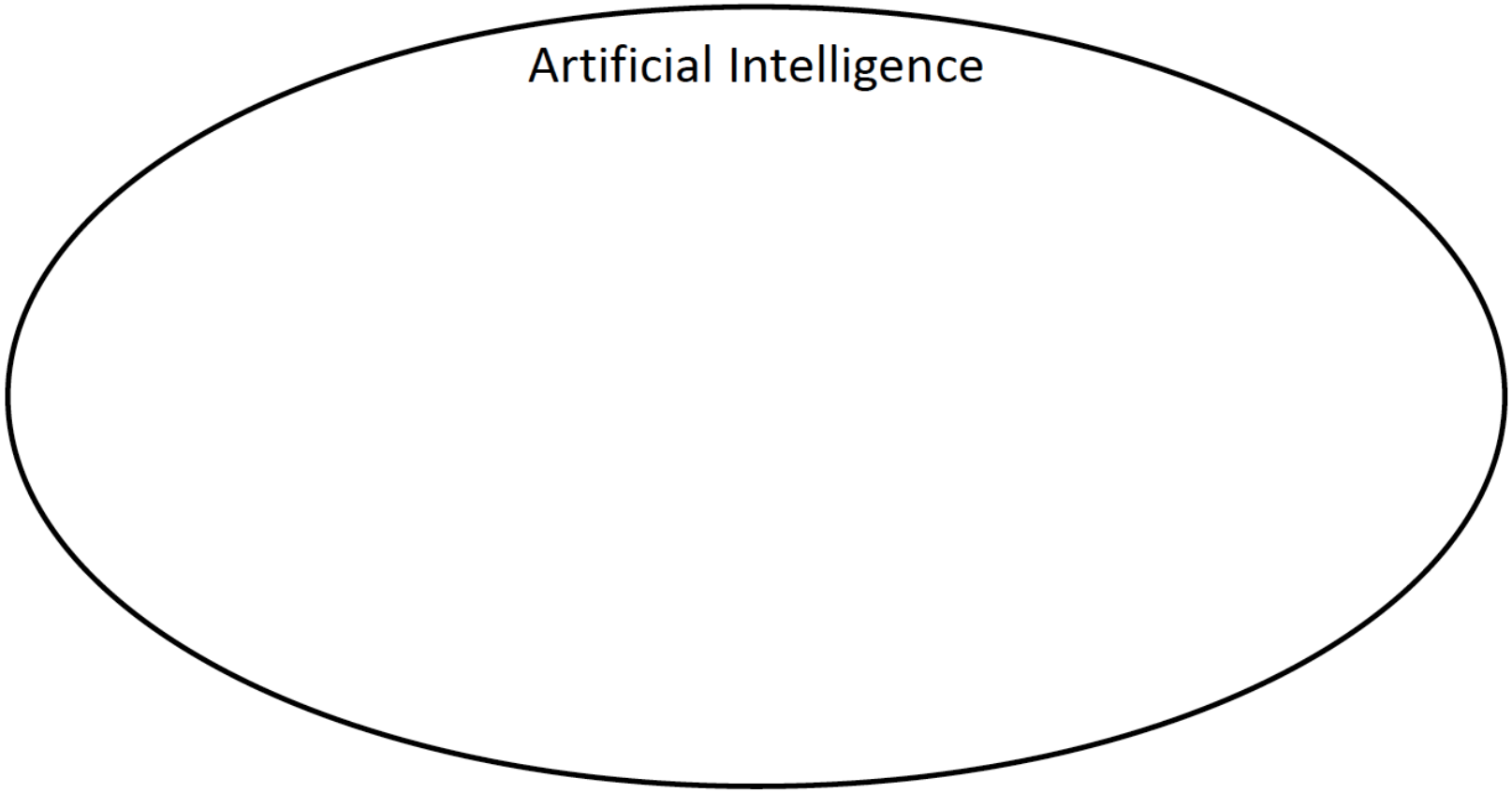
CSE4096 Introduction to Deep Learning

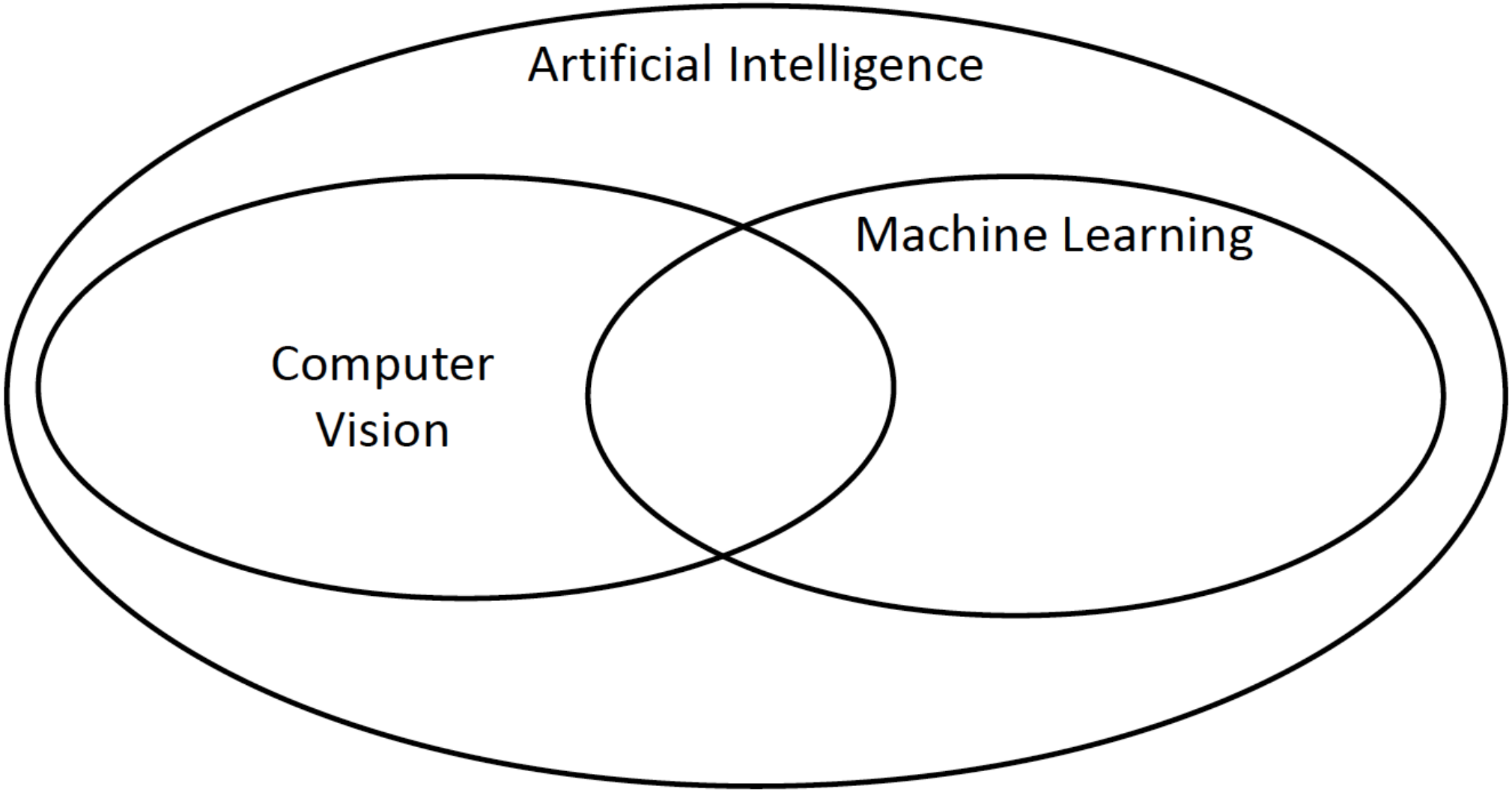
Introduction and Overview

Outline

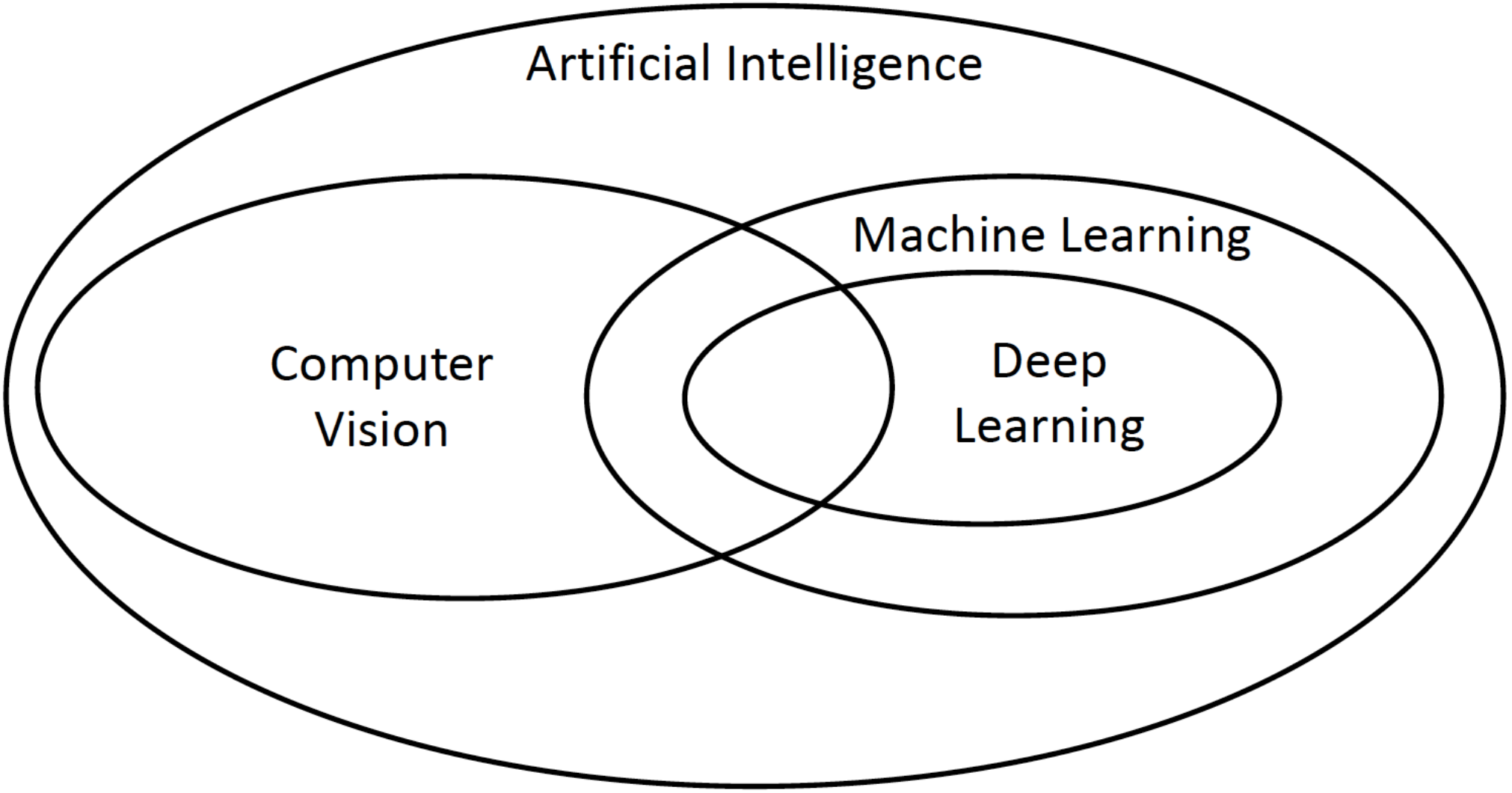
- What is deep learning?
- Brief history of machine learning
- History of computer vision and deep learning
- Why is deep learning so popular?
- What enabled deep learning?
- Applications of deep learning
- Strengths and weaknesses of deep learning

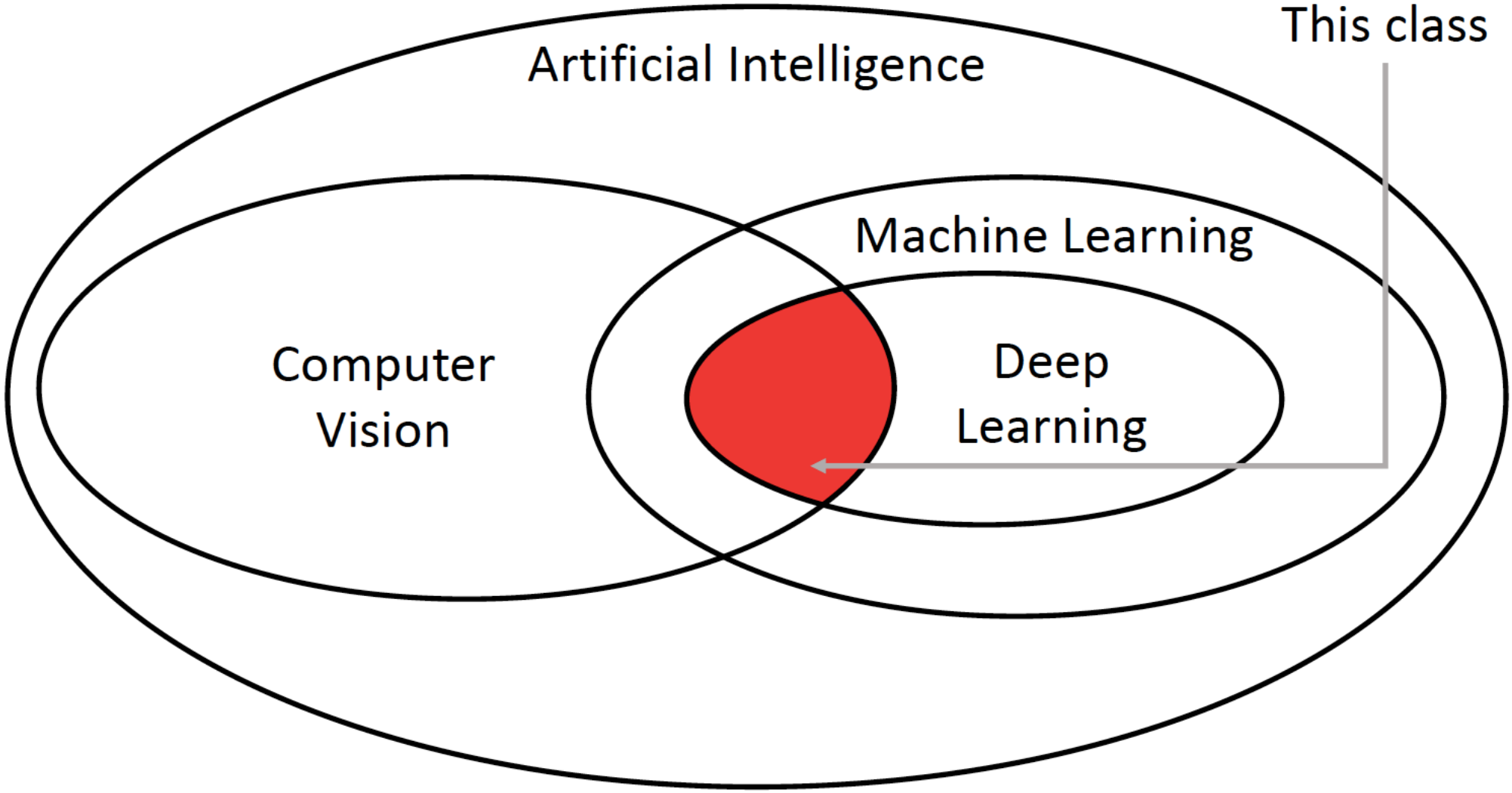


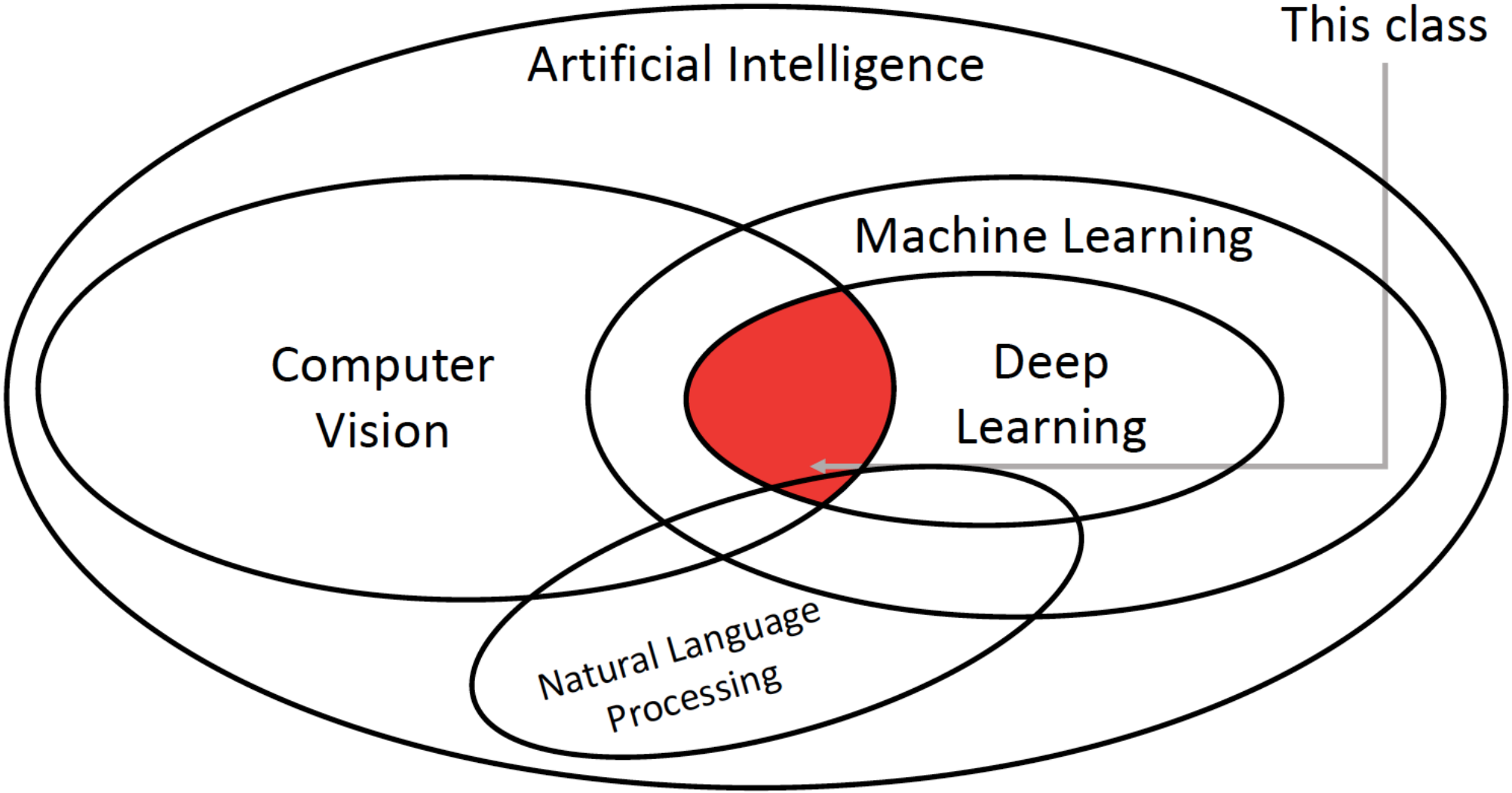
AI can be described as the effort to automate intellectual tasks normally performed by humans. AI is a general field that encompasses machine learning and deep learning, but that also includes many more approaches that may not involve any learning.

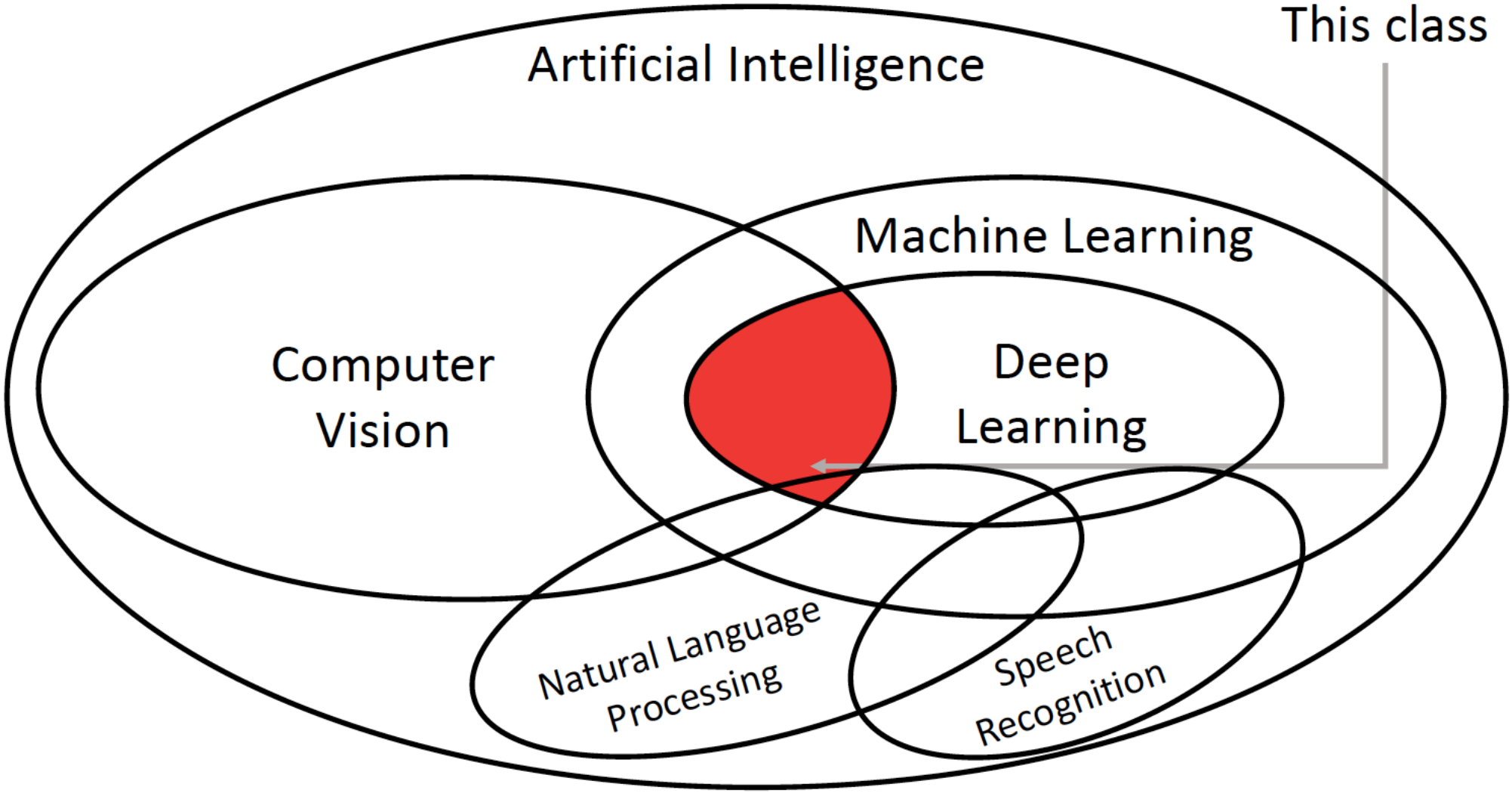


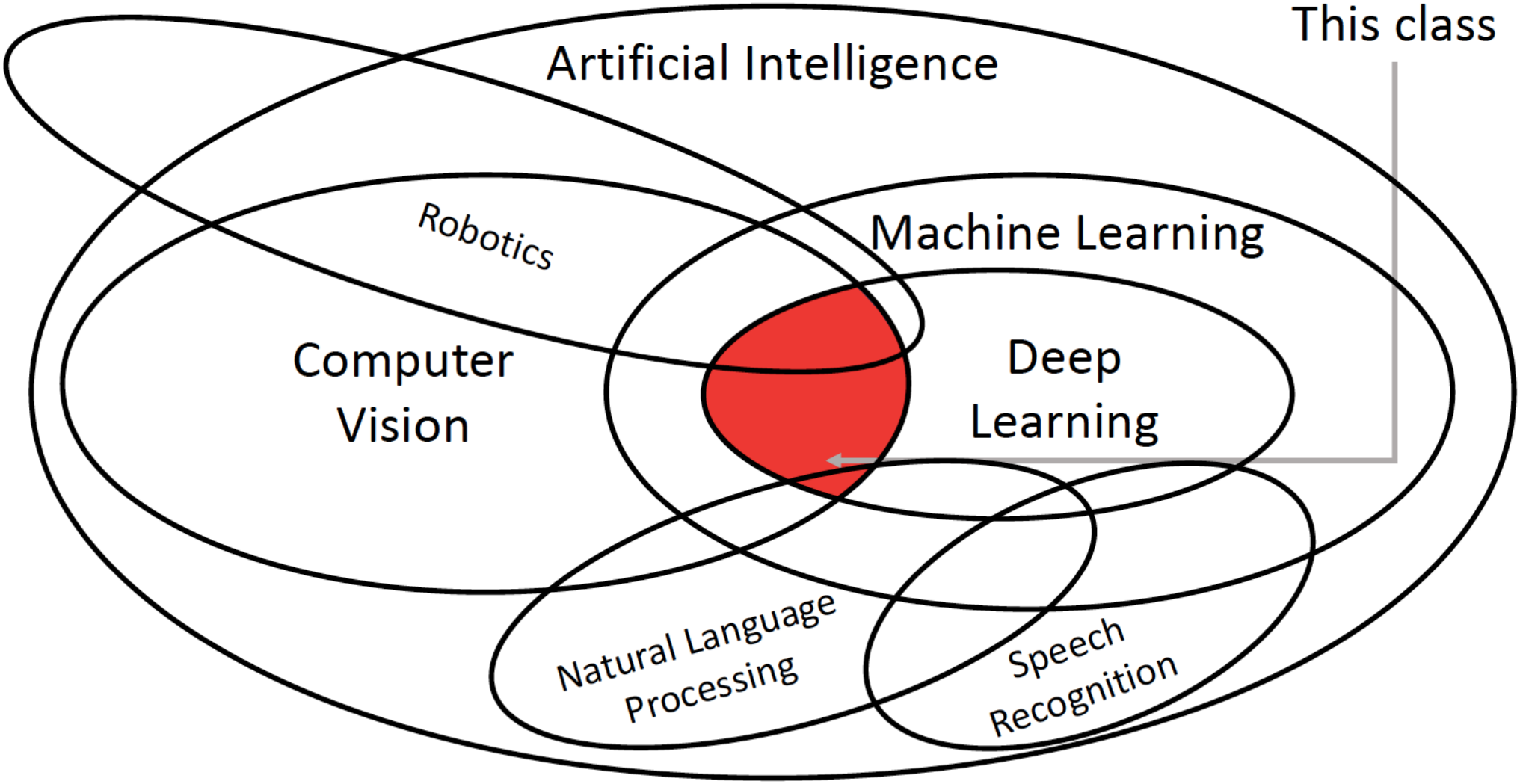
Computer vision and machine learning are the two important fields in AI.


Artificial Intelligence
Machine Learning
Computer
Vision
Deep
Learning


This class
Artificial Intelligence
Machine Learning
Computer
Vision
Deep
Learning


This class
Artificial Intelligence
Machine Learning
Computer
Vision
Deep
Learning
Natural Language
Processing


This class
Artificial Intelligence
Machine Learning
Computer
Vision
Deep
Learning
Natural Language
Processing
Speech
Recognition


This class
Artificial Intelligence
Robotics
Machine Learning
Computer Vision
Deep Learning
Natural Language Processing
Speech Recognition

Deep Learning (DL)

- DL is a branch of **machine learning.**
- **Machine learning:**

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E."

"Machine Learning",Tom Mitchell, McGraw Hill, 1997.

Deep Learning (DL)

- DL is a branch of **machine learning.**
- **Machine learning:**

e.g. categorization

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E."

"Machine Learning",Tom Mitchell, McGraw Hill, 1997.

Deep Learning (DL)

- DL is a branch of **machine learning.**
- **Machine learning:**

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E."

e.g. categorization

Ratio of correctly categorized examples

"Machine Learning",Tom Mitchell, McGraw Hill, 1997.

Deep Learning (DL)

- DL is a branch of **machine learning.**
- **Machine learning:**

Dataset (set of examples)

e.g. categorization

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E."

Ratio of correctly categorized examples

"Machine Learning", Tom Mitchell, McGraw Hill, 1997.

Machine Learning

- A machine learning system is **trained** rather than **explicitly programmed**.
- It's presented with many examples relevant to a task, and it finds statistical structure in these examples that eventually allows the system to come up with rules for automating the task.

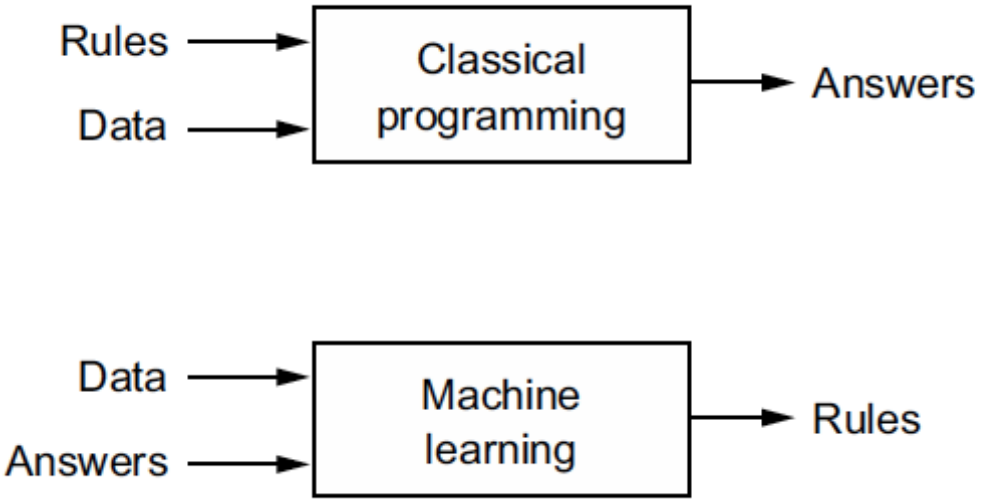


Figure 1.2 Machine learning: a new programming paradigm

Learning rules and representations from data

- A machine learning model transforms its input data into meaningful outputs, a process that is "learned" from exposure to known examples of inputs and outputs.
- Therefore, the central problem in machine learning and deep learning is to *meaningfully transform data*: in other words, to learn useful representations of the input data at hand—**representations** that get us closer to the expected output.

What is a representation?

- It's a different way to look at data—to represent or encode data.
  - For instance, a color image can be encoded in the RGB format (red-green-blue) or in the HSV format (hue-saturation-value): these are two different representations of the same data.
- Some tasks that may be difficult with one representation can become easy with another.
  - For example, the task "select all red pixels in the image" is simpler in the RGB format, whereas "make the image less saturated" is simpler in the HSV format.
- Machine learning models are all about finding appropriate representations for their input data—transformations of the data that make it more amenable to the task at hand.

Representation: Example

- We want to develop an algorithm that can take the coordinates (x, y) of a point and output whether that point is likely to be black or to be white.
  - The inputs are the coordinates of our points.
  - The expected outputs are the colors of our points.
  - A way to measure whether our algorithm is doing a good job could be, for instance, the percentage of points that are being correctly classified.

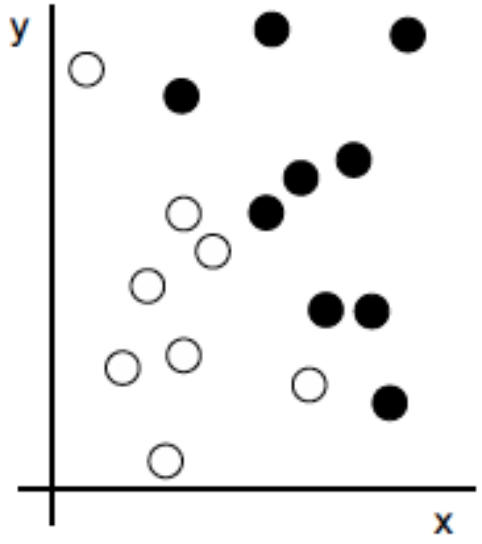
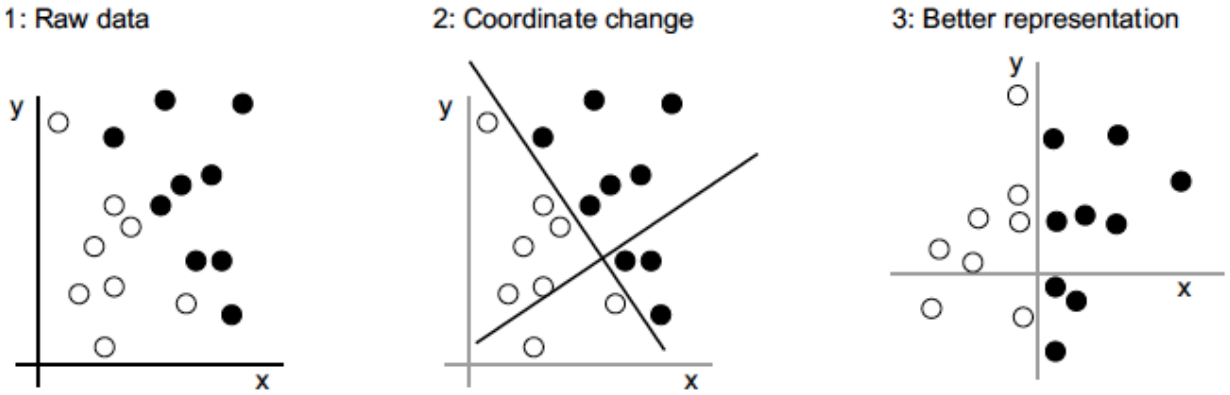


Figure 1.3 Some sample data

Representation: Example

- What we need here is a new representation of our data that cleanly separates the white points from the black points.
- One transformation we could use, among many other possibilities, would be a coordinate change. In this new coordinate system, the coordinates of our points can be said to be a new representation of our data.
- With this representation, the black/white classification problem can be expressed as a simple rule: "Black points are such that $x > 0$," or "White points are such that $x < 0$."
- This new representation, combined with this simple rule, neatly solves the classification problem.



Representations from data

- In this case we defined the coordinate change by hand: we used our human intelligence to come up with our own appropriate representation of the data.
- This is fine for such an extremely simple problem, but could you do the same if the task were to classify images of handwritten digits?
- Could you write down explicit, computer-executable image transformations that would illuminate the difference between a 6 and an 8, between a 1 and a 7, across all kinds of different handwriting?

Representations from data

- This is possible to an extent. Rules based on representations of digits such as “number of closed loops” or vertical and horizontal pixel histograms can do a decent job of telling apart handwritten digits.
- But finding such useful representations by hand is hard work, and, as you can imagine, the resulting rule-based system is brittle— a nightmare to maintain.
- Every time you come across a new example of handwriting that breaks your carefully thought-out rules, you will have to add new data transformations and new rules, while taking into account their interaction with every previous rule.

Representations from data

- If this process is so painful, could we automate it?
- What if we tried systematically searching for different sets of automatically generated representations of the data and rules based on them, identifying good ones by using as feedback the percentage of digits being correctly classified in some development dataset?
- We would then be doing machine learning.
- Learning, in the context of machine learning, describes an automatic search process for data transformations that produce useful representations of some data, guided by some feedback signal—representations that are amenable to simpler rules solving the task at hand.

Representations from data

- These transformations can be:
  - coordinate changes (like in our 2D coordinates classification example),
  - taking a histogram of pixels and counting loops (like in our digits classification example),
  - they could also be linear projections, translations,
  - nonlinear operations (such as “select all points such that $x > 0$”), and so on.

Representations from data

- Machine learning algorithms aren't usually creative in finding these transformations;
  - they're merely searching through a predefined set of operations, called a hypothesis space.
  - For instance, the space of all possible coordinate changes would be our hypothesis space in the 2D coordinates classification example.

Machine Learning

- What machine learning is, concisely:
  - searching for useful representations and rules over some input data,
  - within a predefined space of possibilities,
  - using guidance from a feedback signal.

Deep Learning

- A new way of learning representations from data that puts an emphasis on learning successive layers of increasingly meaningful representations.
- The "deep" in "**deep learning**" stands for this idea of **successive layers of representations**.
- How many layers contribute to a model of the data is called the **depth** of the model.
- Other appropriate names for the field could have been layered representations learning or hierarchical representations learning.
- Other approaches to machine learning tend to focus on learning only **one or two layers of representations** of the data (say, taking a pixel histogram and then applying a classification rule); hence, they're sometimes called **shallow learning**.
- Classical machine learning methods (SVM, decision trees etc.) requires first extracting useful representations manually (a step called **feature engineering**), which is difficult and brittle.

Deep Learning (DL)

DL: the set of algorithms and models that work on multilayer graphs called artificial neural networks which were inspired by the structure and function found in the biological brain.

In a DL model, multiple layers of linear and non-linear operations are carried out.

“Deep” implies more than a few layers (between input and output).

$$\hat{\mathbf{y}} = f(\mathbf{x}; \boldsymbol{\Theta})$$

Input

Output

Model

Parameters

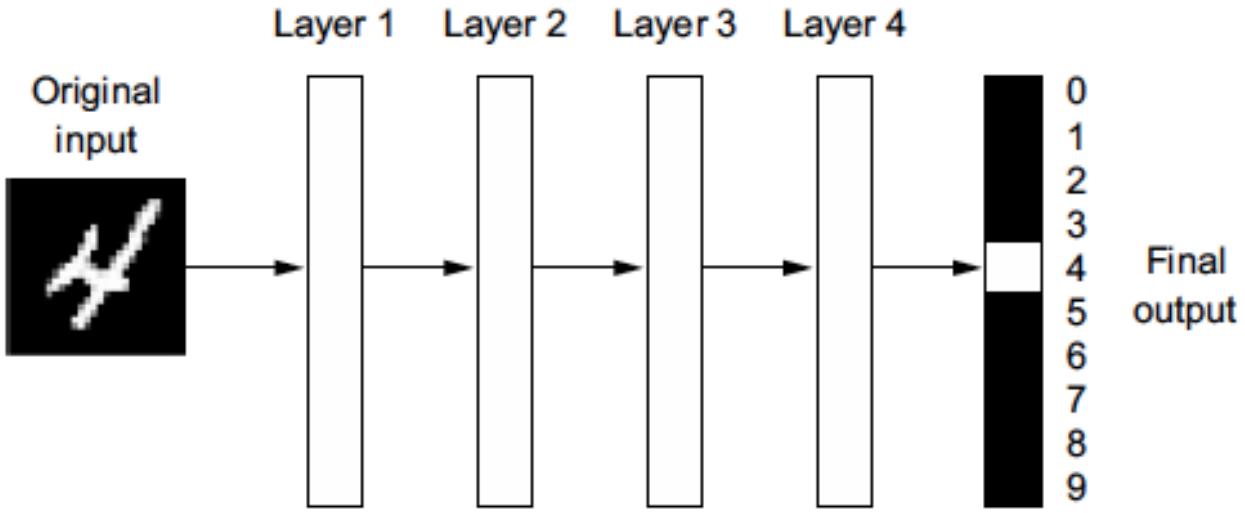
Task

$$\hat{\mathbf{y}} = f(\mathbf{x}; \boldsymbol{\Theta})$$

$$\hat{\mathbf{y}} = g_N(...g_3(g_2(g_1(\mathbf{x}; \boldsymbol{\Theta_1}); \boldsymbol{\Theta_2}); \boldsymbol{\Theta_3}); ...\boldsymbol{\Theta_N})$$

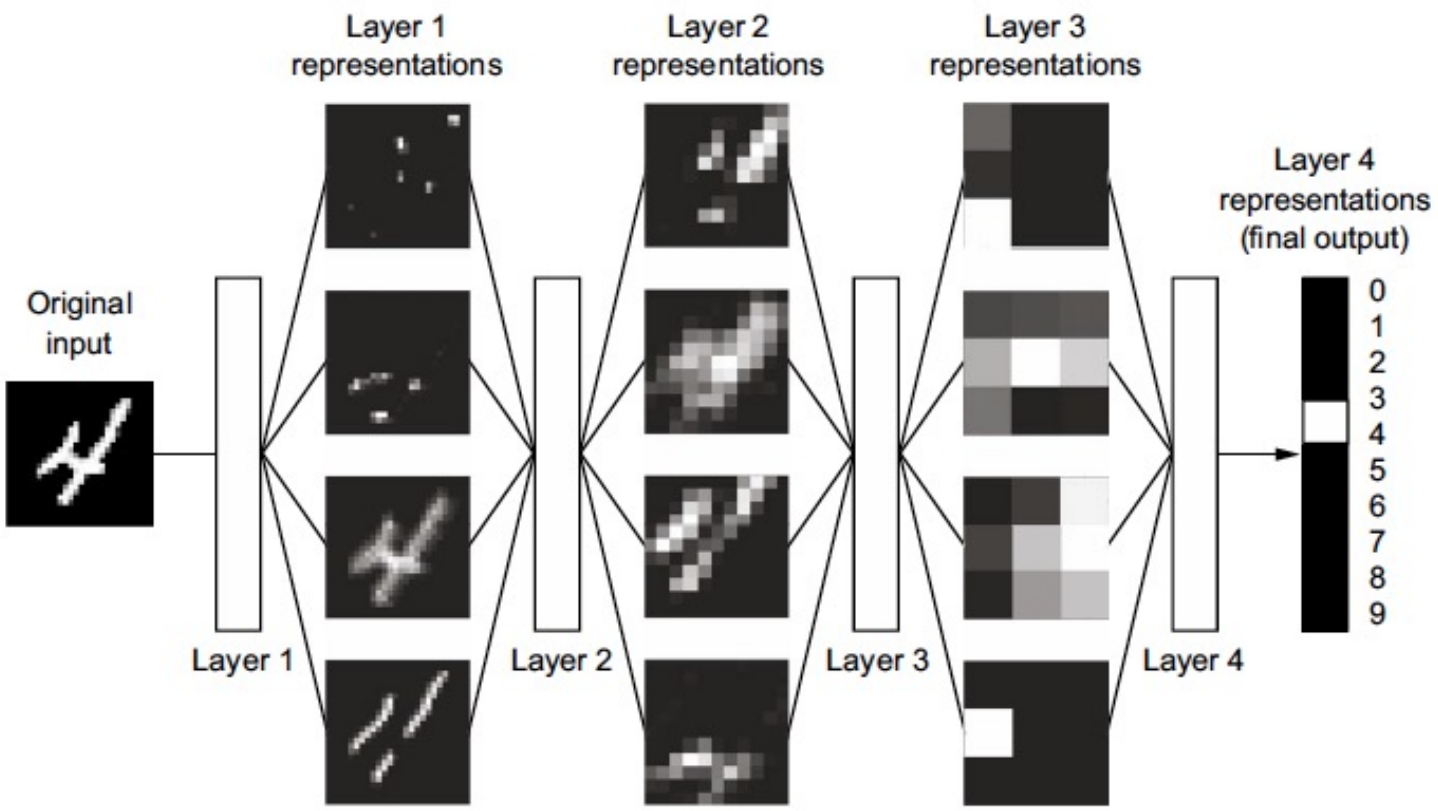
Representations Learned By Deep Learning

- Let's examine how a network several layers deep transforms an image of a digit in order to recognize what digit it is.

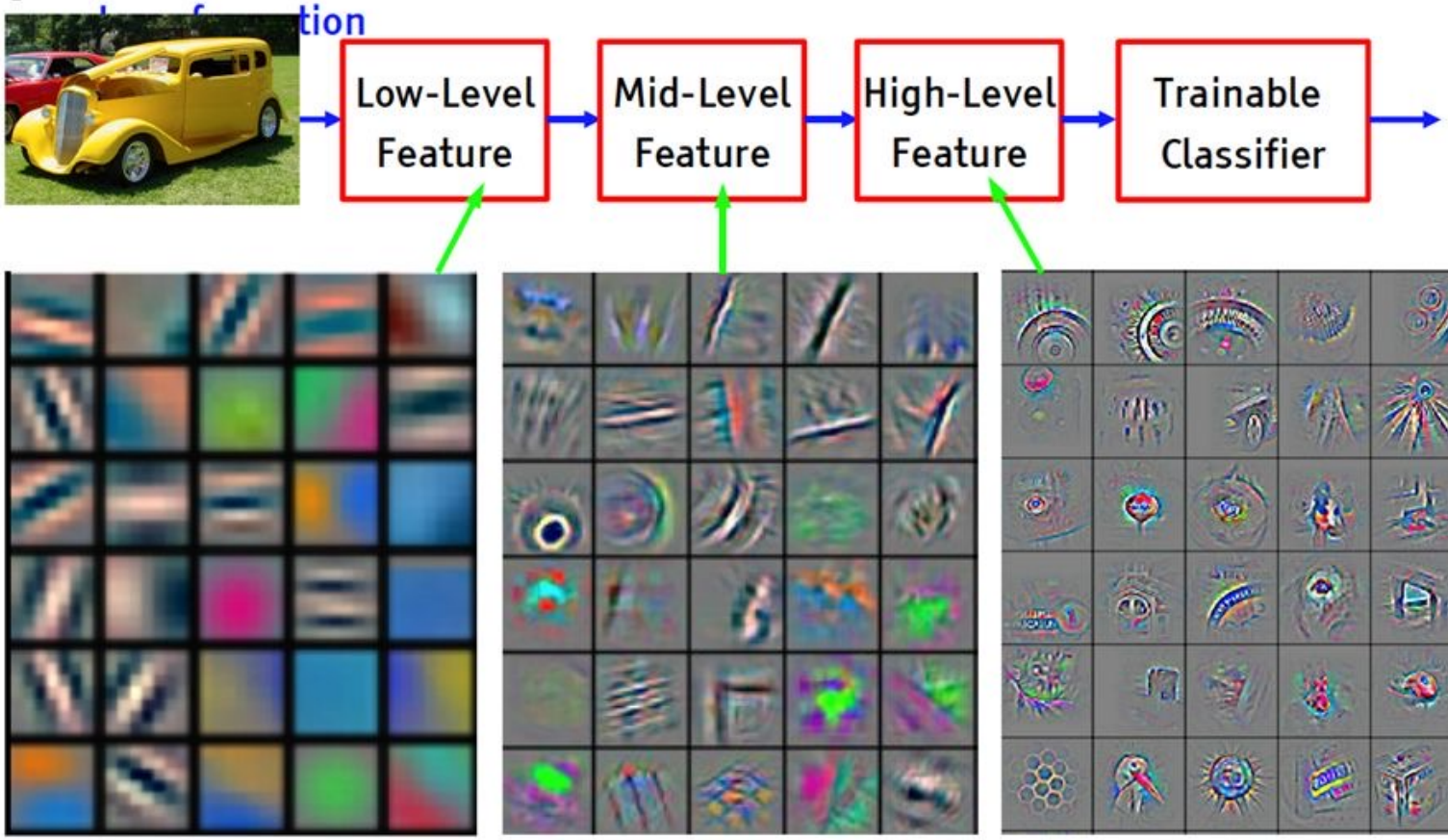


Representations Learned By Deep Learning

- The network transforms the digit image into representations that are increasingly different from the original image and increasingly informative about the final result.
- A deep network as a multistage **information distillation** process, where information goes through successive filters and comes out increasingly purified (that is, useful with regard tc some task).
- Deep learning is a multistage way to learn data representations.

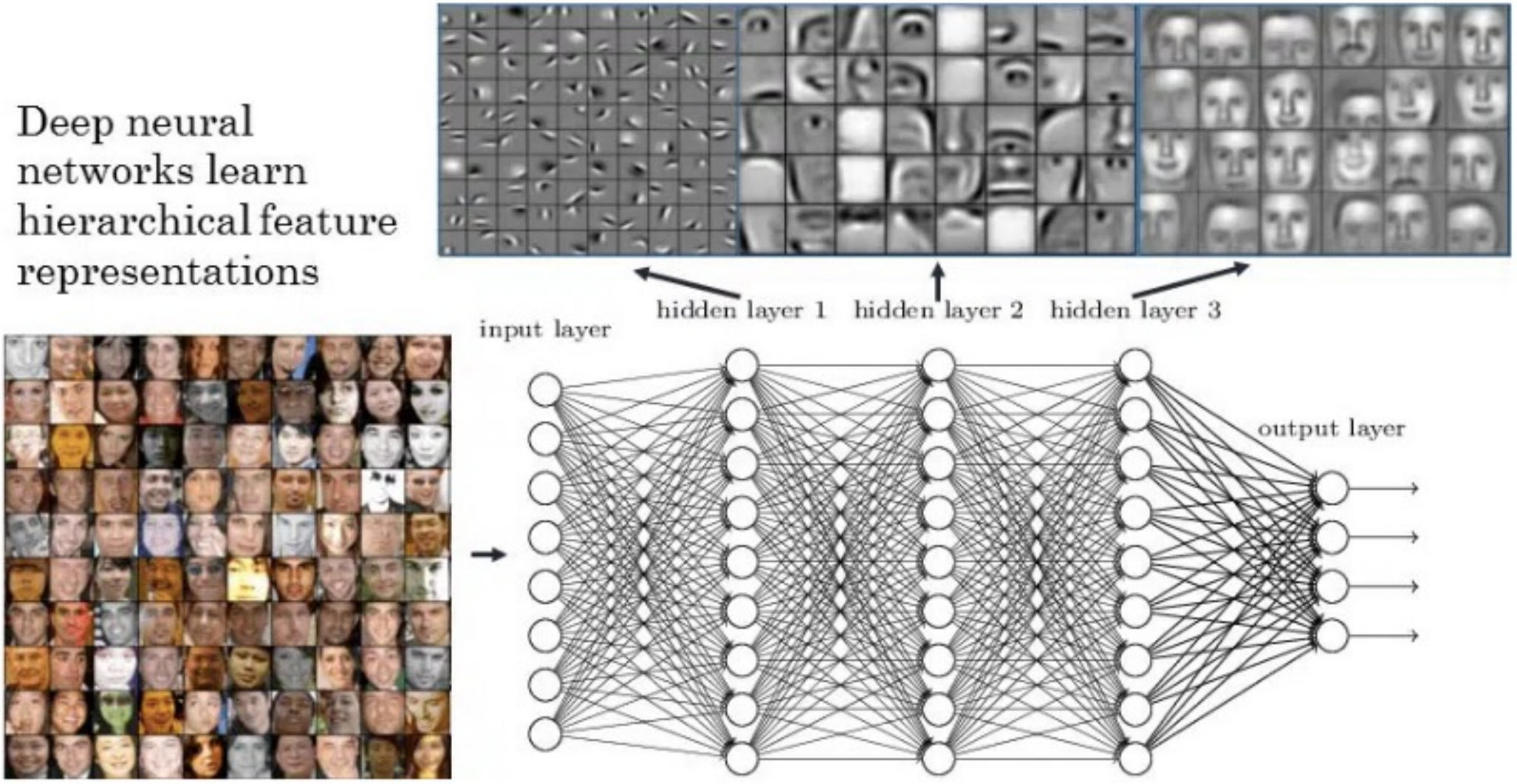


Deep Learning = Learning Hierarchical Representations



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

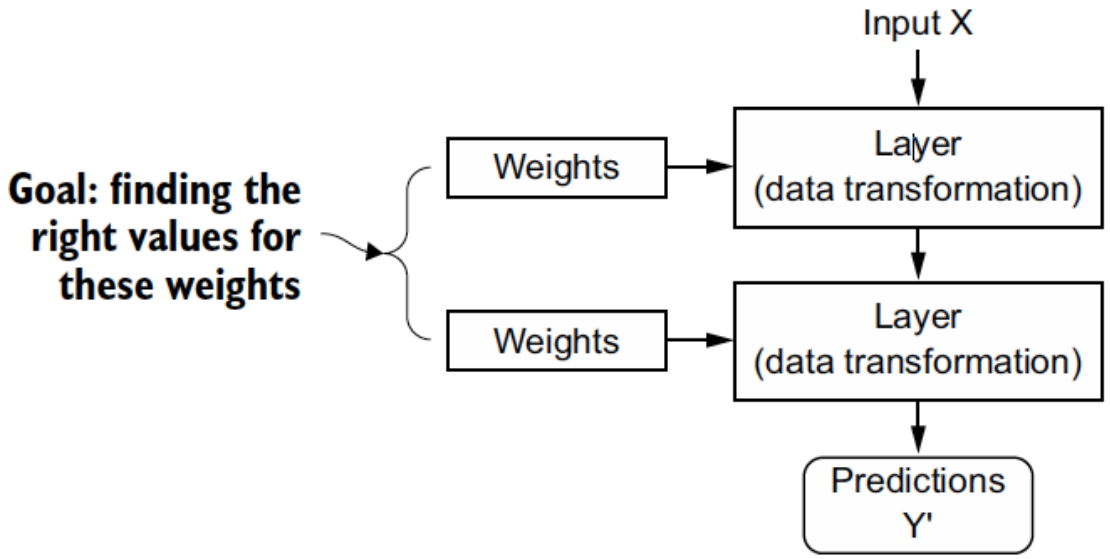
Deep neural networks learn hierarchical feature representations



https://www.rsipvision.com/exploring-deep-learning/

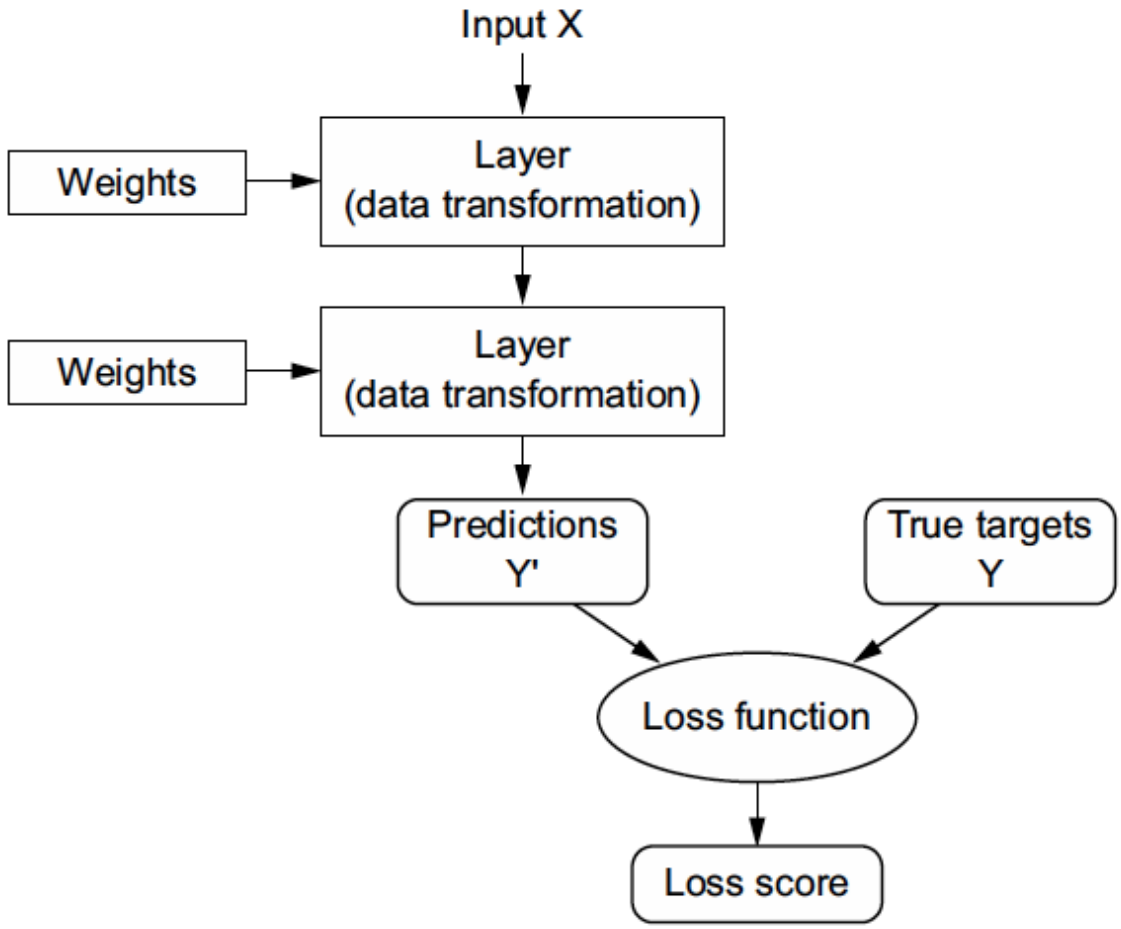
How does deep learning work?

- What a layer does to its input data is stored in the layer's **weights**, which in essence are a bunch of numbers.
- The transformation implemented by a layer is parameterized by its **weights (parameters** of a layer.)
- Learning means finding a set of values for the weights of all layers in a network, such that the network will correctly map example inputs to their associated targets.
- A deep neural network can contain tens of millions of parameters!



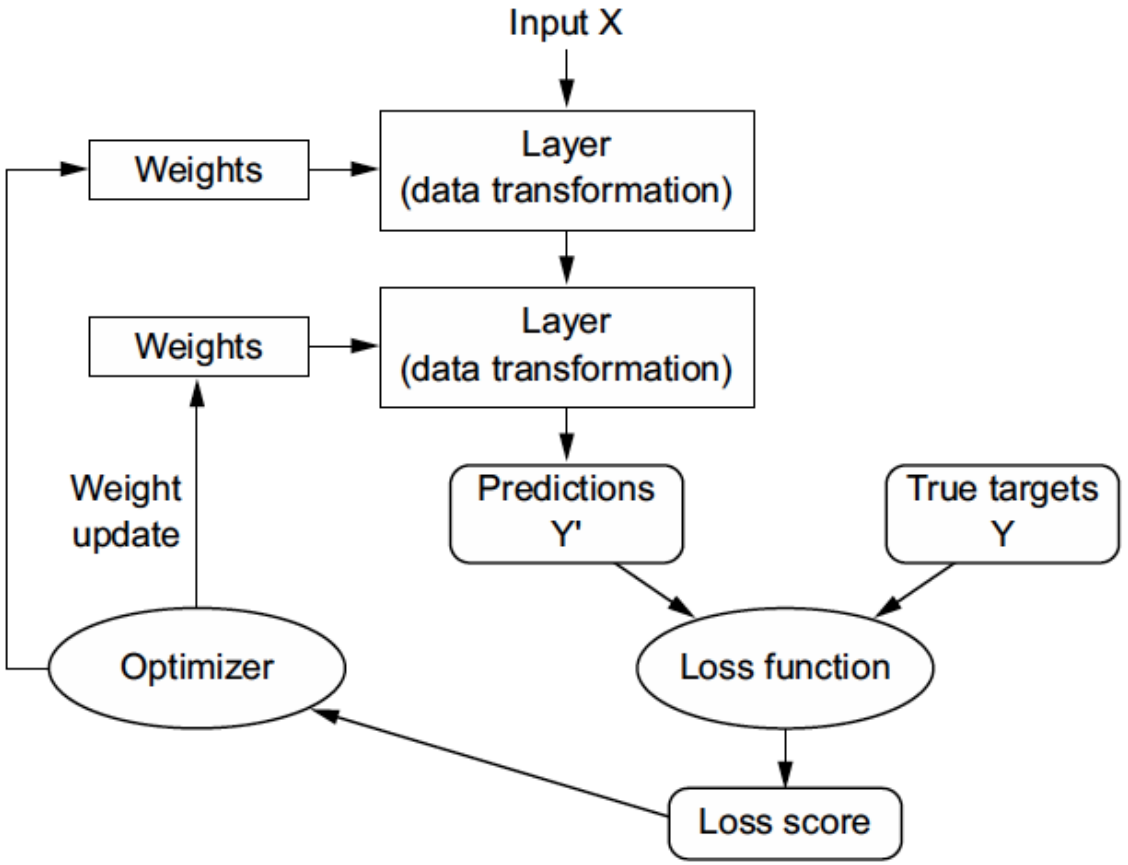
How does deep learning work?

- To control the output of a neural network, you need to be able to measure how far this output is from what you expected.
- This is the job of the **loss function** of the network, also sometimes called the **objective function** or **cost function**.
- The loss function takes the predictions of the network and the true target (what you wanted the network to output) and computes a distance score, capturing how well the network has done on this specific example.



How does deep learning work?

- The fundamental trick in deep learning is to use this score as a feedback signal to adjust the value of the weights a little, in a direction that will lower the loss score for the current example.
- This adjustment is the job of the **optimizer**, which implements what's called the **Backpropagation** algorithm: the central algorithm in deep learning.



How does deep learning work?

- Initially, the weights of the network are assigned random values, so the network merely implements a series of random transformations.
- Naturally, its output is far from what it should ideally be, and the loss score is accordingly very high.
- But with every example the network processes, the weights are adjusted a little in the correct direction, and the loss score decreases.
- This is the **training loop**, which, repeated a sufficient number of times (typically tens of iterations over thousands of examples), yields weight values that minimize the loss function.
- A network with a minimal loss is one for which the outputs are as close as they can be to the targets: **a trained network**.

A BRIEF HISTORY OF MACHINE LEARNING

Machine Learning

- It's safe to say that most of the machine learning algorithms used in the industry today aren't deep learning algorithms.
- Deep learning isn't always the right tool for the job:
  - Sometimes there isn't enough data for deep learning to be applicable,
  - Sometimes the problem is better solved by a different algorithm.
- If deep learning is your first contact with machine learning, you may find yourself in a situation where all you have is the deep learning hammer, and every machine learning problem starts to look like a nail.
- The only way not to fall into this trap is to be familiar with other approaches and practice them when appropriate.

ML: Probabilistic Modeling

- It is one of the earliest forms of machine learning, and it's still widely used to this day.
- Best-known algorithms:
  - **Naive Bayes algorithm**:
    - Based on applying Bayes' theorem
    - Assumes that the features in the input data are all independent (a strong, or "naive" assumption, which is where the name comes from).
  - **Logistic Regression**:
    - Is a classification algorithm.
    - "Hello World" of modern machine learning.
    - Often the first thing a data scientist will try on a dataset to get a feel for the classification task at hand.

ML: Early Neural Networks

- Although the core ideas of neural networks were investigated in toy forms as early as the **1950s**, the approach took decades to get started.
- For a long time, the missing piece was an efficient way to train large neural networks.
- This changed in the **mid-1980s**, when multiple people independently rediscovered the Backpropagation algorithm— a way to train chains of parametric operations using gradient-descent optimization and started applying it to neural networks.
- The first successful practical application of neural nets came in **1989** from Bell Labs, when Yann LeCun combined the earlier ideas of convolutional neural networks and backpropagation and applied them to the problem of classifying handwritten digits.
- The resulting network, dubbed LeNet, was used by the United States Postal Service in the **1990**s to automate the reading of ZIP codes on mail envelopes.

ML: Kernel Methods, Support Vector Machines (SVM)

- Kernel methods are a group of classification algorithms, the best known of which is SVM.
- The modern formulation of an SVM was developed by Vladimir Vapnik and Corinna Cortes in the early 1990s at Bell Labs and published in 1995, although an older linear formulation was published by Vapnik and Alexey Chervonenkis as early as 1963.
- SVM is a classification algorithm that works by finding "**decision boundaries**" separating two classes.
- SVMs proceed to find these boundaries in two steps:
  - The **data is mapped to a new high-dimensional representation where the decision boundary can be expressed as a hyperplane** (if the data was two-dimensional, a hyperplane would be a straight line).
  - A good decision boundary (a separation hyperplane) is computed by trying to maximize the distance between the hyperplane and the closest data points from each class, a step called maximizing the margin. This allows the boundary to generalize well to new samples outside of the training dataset.

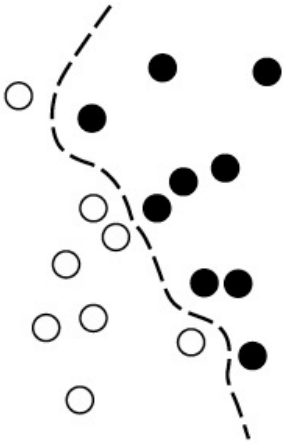


Figure 1.10
A decision boundary

SVM - Kernel Trick

- The technique of mapping data to a high-dimensional representation where a classification problem becomes simpler is possible by **kernel trick.**
- **Kernel trick:** To find good decision hyperplanes in the new representation space, you don't have to explicitly compute the coordinates of your points in the new space; you just need to compute the distance between pairs of points in that space, which can be done efficiently using a kernel function.
- **A kernel function** is a computationally tractable operation that maps any two points in your initial space to the distance between these points in your target representation space, completely bypassing the explicit computation of the new representation.
- Kernel functions are typically crafted by hand rather than learned from data—in the case of an SVM, only the separation hyperplane is learned.

SVM - Drawbacks

- At the time they were developed, SVMs exhibited state-of-the-art performance on simple classification problems and were one of the few machine learning methods backed by extensive theory and amenable to serious mathematical analysis, making them well understood and easily interpretable.
- Because of these useful properties, SVMs became extremely popular in the field for a long time.
- But SVMs proved **hard to scale to large datasets** and **didn't provide good results for perceptual problems such as image classification**.
- Because an SVM is a shallow method, applying an SVM to perceptual problems **requires first extracting useful representations manually** (a step called feature engineering), which is difficult and brittle.
  - For instance, if you want to use an SVM to classify handwritten digits, you can't start from the raw pixels; you should first find by hand useful representations that make the problem more tractable, like the pixel histograms mentioned earlier.

ML: Decision Trees, Random Forests, Gradient Boosting Machines

- **Decision trees** are flowchart-like structures that let you classify input data points or predict output values given inputs.
- They're easy to visualize and interpret.
- Decision trees learned from data began to receive significant research interest in the 2000s, and by 2010 they were often preferred to kernel methods.

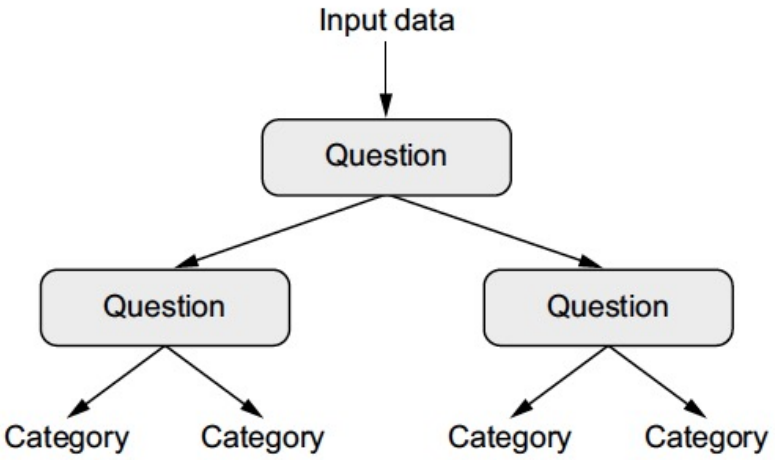


Figure 1.11 A decision tree: the parameters that are learned are the questions about the data. A question could be, for instance, "Is coefficient 2 in the data greater than 3.5?"

ML: Decision Trees, Random Forests, Gradient Boosting Machines

- **Random Forest** algorithm introduced a robust, practical take on decision-tree learning that involves building a large number of specialized decision trees and then ensembling their outputs.
- Random forests are applicable to a wide range of problems—you could say that they're almost always the second-best algorithm for any shallow machine learning task.
- When the popular machine learning competition website
- Kaggle (http://kaggle.com) got started in 2010, random forests quickly became a favorite on the platform—until 2014, when **gradient boosting machines** took over.

ML: Decision Trees, Random Forests, Gradient Boosting Machines

- **A gradient boosting machine** is based on ensembling weak prediction models, generally decision trees.
- It uses gradient boosting, a way to improve any machine learning model by iteratively training new models that specialize in addressing the weak points of the previous models.
- Applied to decision trees, the use of the gradient boosting technique results in models that strictly outperform random forests most of the time, while having similar properties.
- It may be one of the best, if not the best, algorithm for dealing with non-perceptual data today.
- Alongside deep learning, it's one of the most commonly used techniques in Kaggle competitions.

What makes deep learning different

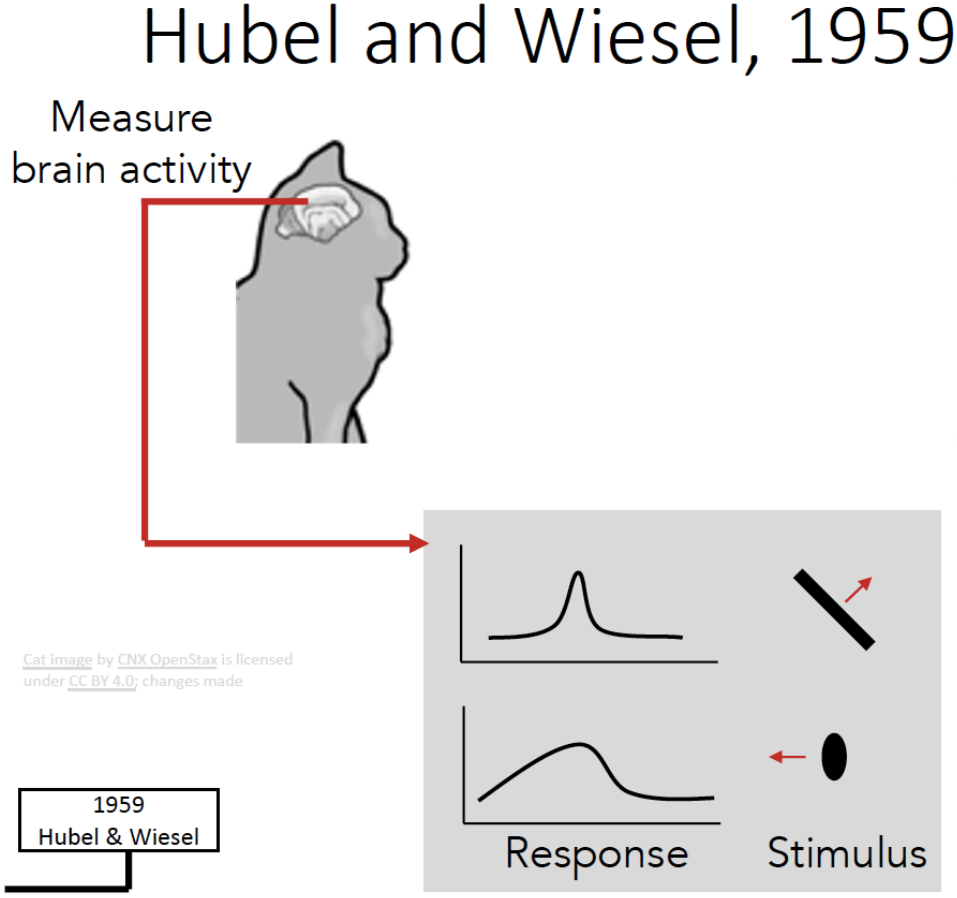
- The primary reason deep learning is dominant today is that it offered better performance for many problems.
- But that's not the only reason. Deep learning also makes problem-solving much easier, because it **completely automates** what used to be the most crucial step in a machine learning workflow: **feature engineering**.
  - Previous machine learning techniques—shallow learning—only involved transforming the input data into one or two successive representation spaces, usually via simple transformations such as high-dimensional non-linear projections (SVMs) or decision trees.
  - But the refined representations required by complex problems generally can't be attained by such techniques.
  - As such, humans had to go to great lengths to make the initial input data more amenable to processing by these methods: they had to manually engineer good layers of representations for their data.
  - This is called feature engineering.

What makes deep learning different

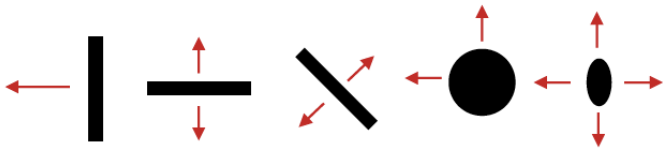
- Deep learning, on the other hand, completely automates this step: with deep learning, you learn all features in one pass rather than having to engineer them yourself.
- What is transformative about deep learning is that it allows a model **to learn all layers of representation jointly**, at the same time, rather than in succession (greedily, as it's called).
- With joint feature learning, whenever the model adjusts one of its internal features, all other features that depend on it automatically adapt to the change, without requiring human intervention.
- Everything is supervised by a single feedback signal: every change in the model serves the end goal.
- These are the two essential characteristics of how deep learning learns from data:
  - the **incremental, layer-by-layer way in which increasingly complex representations** are developed,
  - these intermediate **incremental representations are learned jointly**, each layer being updated to follow both the representational needs of the layer above and the needs of the layer below.

HISTORY OF COMPUTER VISION AND DEEP LEARNING

Hubel and Wiesel, 1959

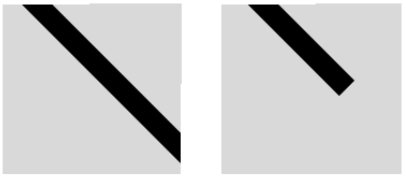


Cat image by CNX OpenStax is licensed under CC BY 4.0; changes made



Simple cells: Response to light orientation

Complex cells: Response to light orientation and movement



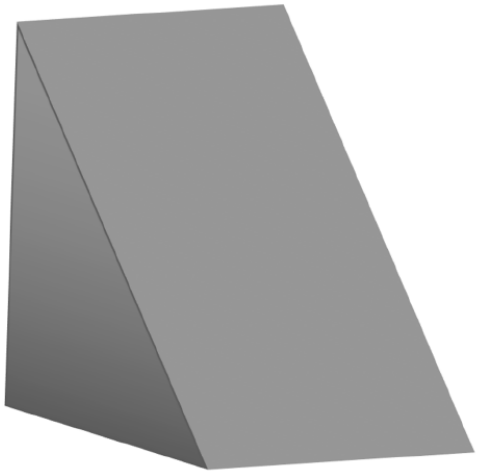
They noticed that the neurons respond at the time when they change the slides. They discovered that it was the moving vertical bar that was causing neural activity in the brain.

Goal: To understand how the visual system works

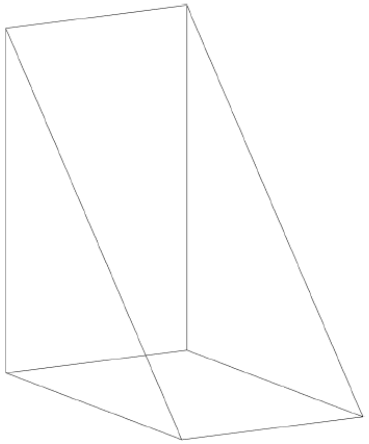
Setup: Show the cat different visual stimuli and record the neural activity using electrodes.

Main Results: There are different neurons that respond to different oriented edges. There is a hierarcy of cells from simple to complex. → Won a Nobel prize in 1981

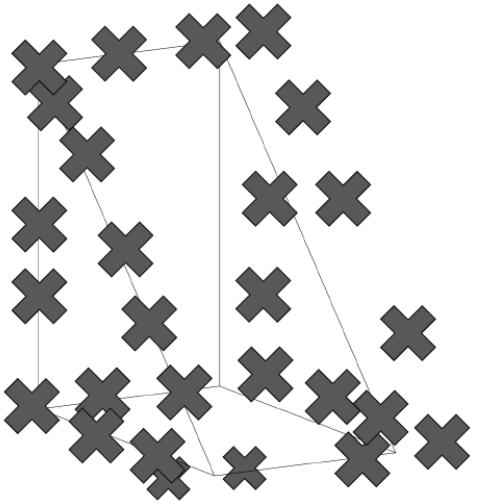
Larry Roberts, 1963



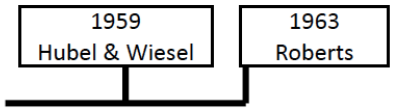
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



Lawrence Gilman Roberts, "Machine Perception of Three-Dimensional Solids", 1963

First PhD thesis on computer vision.

Goal: Undestand the 3D structure of objects.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

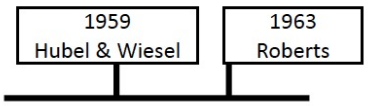
Artificial Intelligence Group — July 7, 1966
Vision Memo. No. 100.

THE SUMMER VISION PROJECT

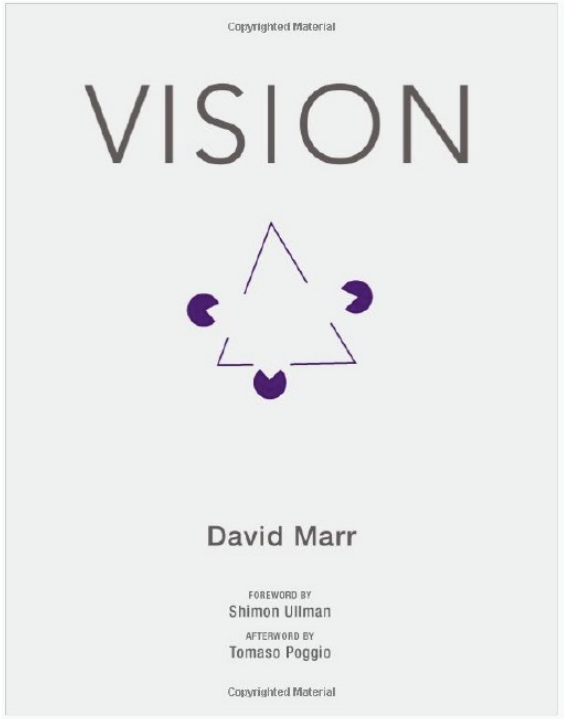
Seymour Papert

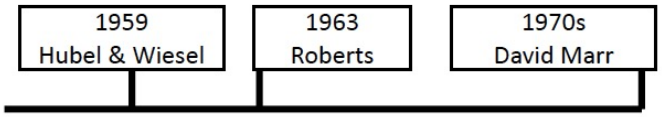
The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

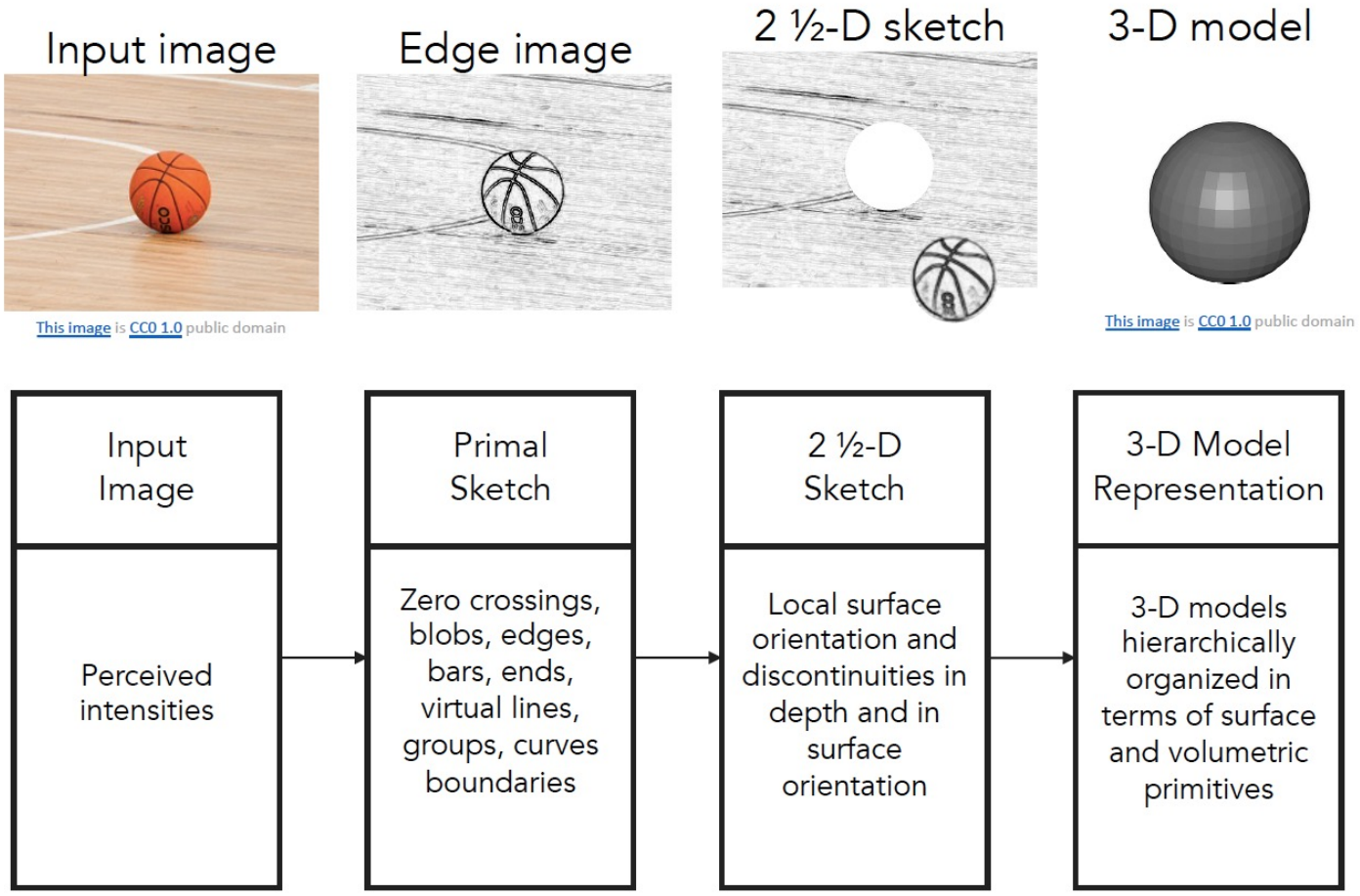
https://dspace.mit.edu/handle/1721.1/6125



Goal: Understand how humans recognize patterns during the summer project. Ambitious goal. Still not solved!



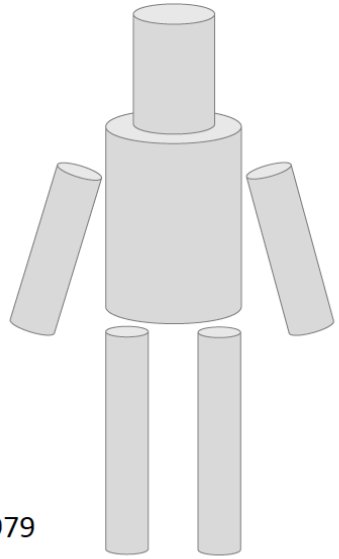




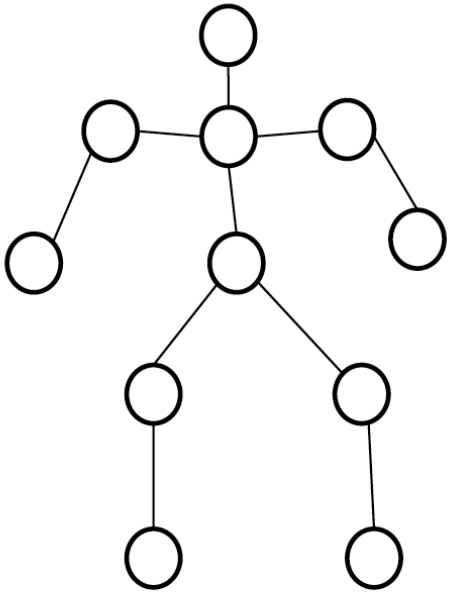
Stages of Visual Representation, David Marr, 1970s

Goal: Segment the objects and estimate the depths of them to come up with a 3D model of the world.

Recognition via Parts (1970s)



Generalized Cylinders,
Brooks and Binford, 1979



Pictorial Structures,
Fischler and Elshlager, 1973

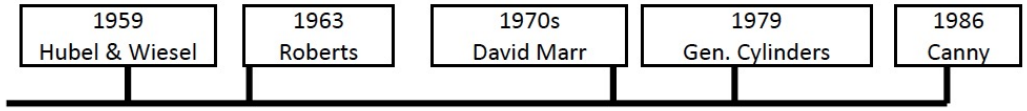


Goal: Instead of using simple structures like edges, try to model more complex objects like people with some known topology. Images and computers were not widely available.

Recognition via Edge Detection (1980s)







John Canny, 1986
David Lowe, 1987

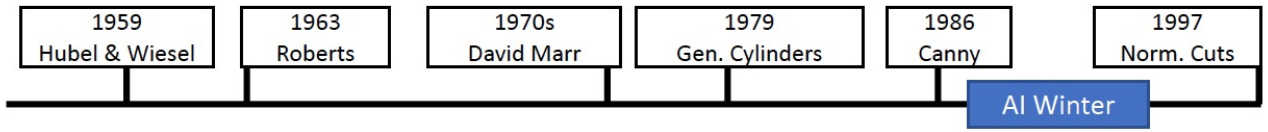
Image is CC0 1.0 public domain

Realistic images and computers were more available.

Goal: Recognize objects using edges: match a template razor in the edge image.

Recognition via Grouping (1990s)





Normalized Cuts, Shi and Malik, 1997

Left Image is CC BY 3.0 Middl Image is public domain Right Image is CC-BY 2.0; changes made

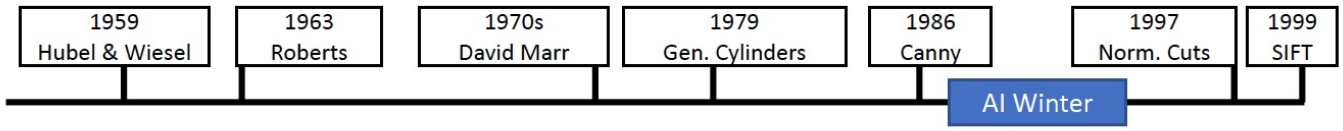
Goal: Recognize objects in more realistic images. Segment the images into semantically meaningful regions which will lead to objects.

Recognition via Matching (2000s)



Image is public domain

Image is public domain



SIFT, David Lowe, 1999

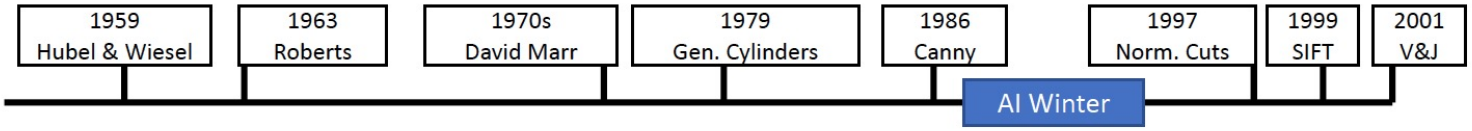
Detect recognizable “key points” in the image and represent its appearance using a feature vector which are invariant to rotations and illumination variations. Establish correspondance between keypoints of two images of the same object to recognize the object.

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision

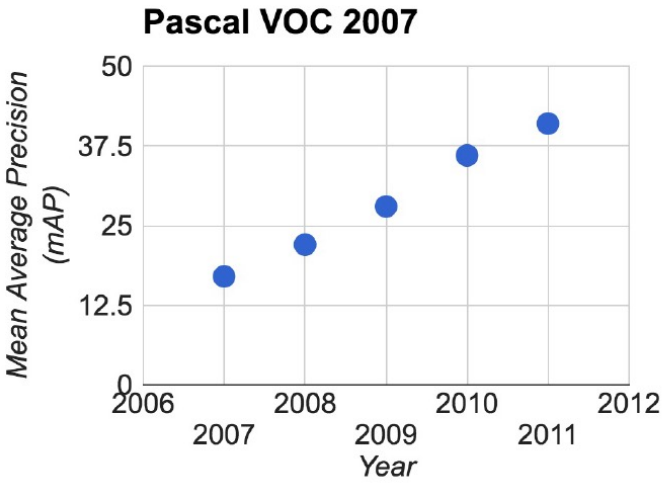


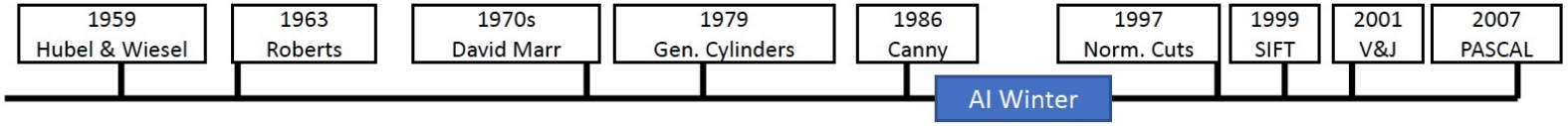


They used boosted decision trees to detect faces. First **major use of machine learning** in computer vision. This algorithm was **used in digital cameras** to detect faces and autofocus on faces.

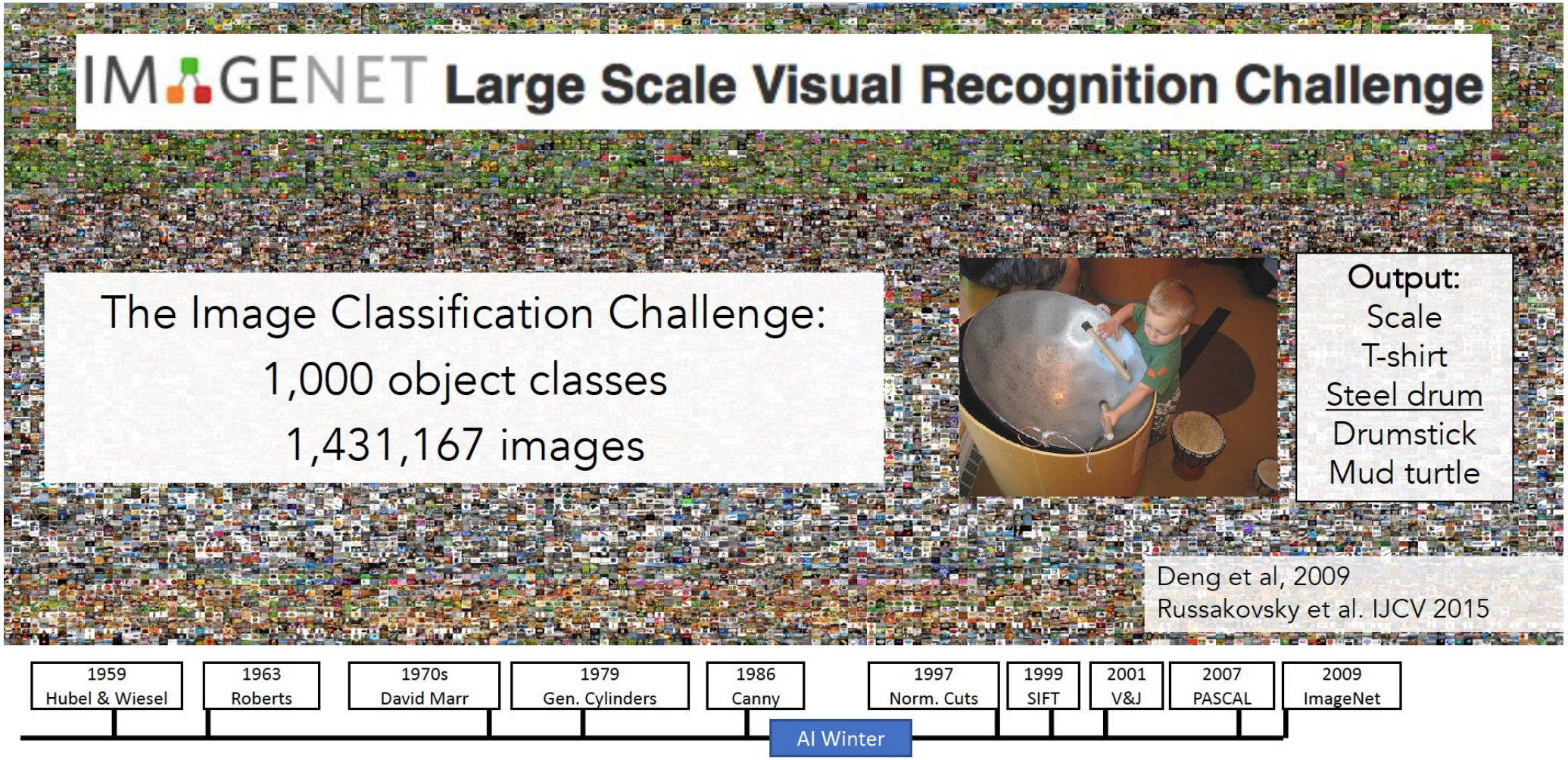
PASCAL Visual Object Challenge



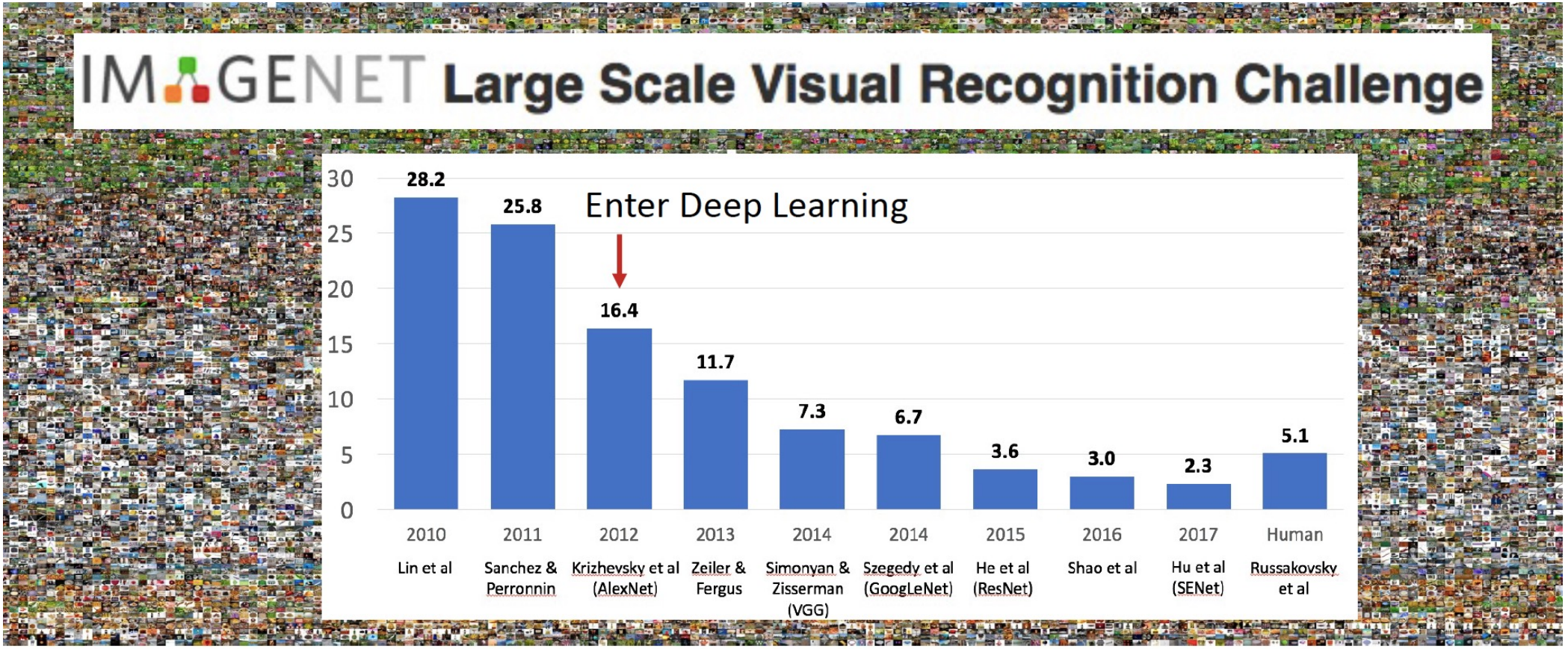


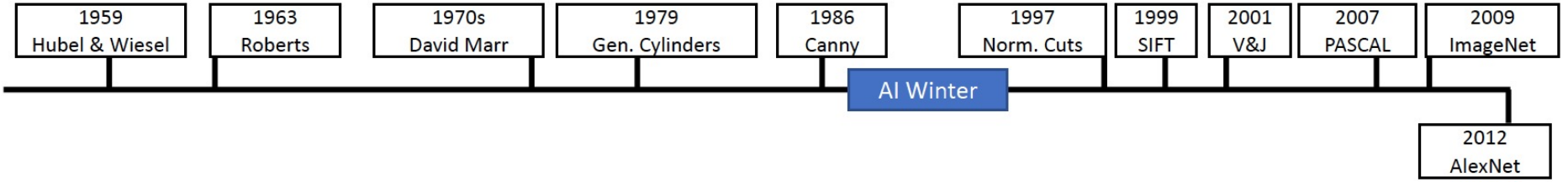


More and more data and machine learning was used in 2000's. Images could be downloaded from the internet to build large datasets.



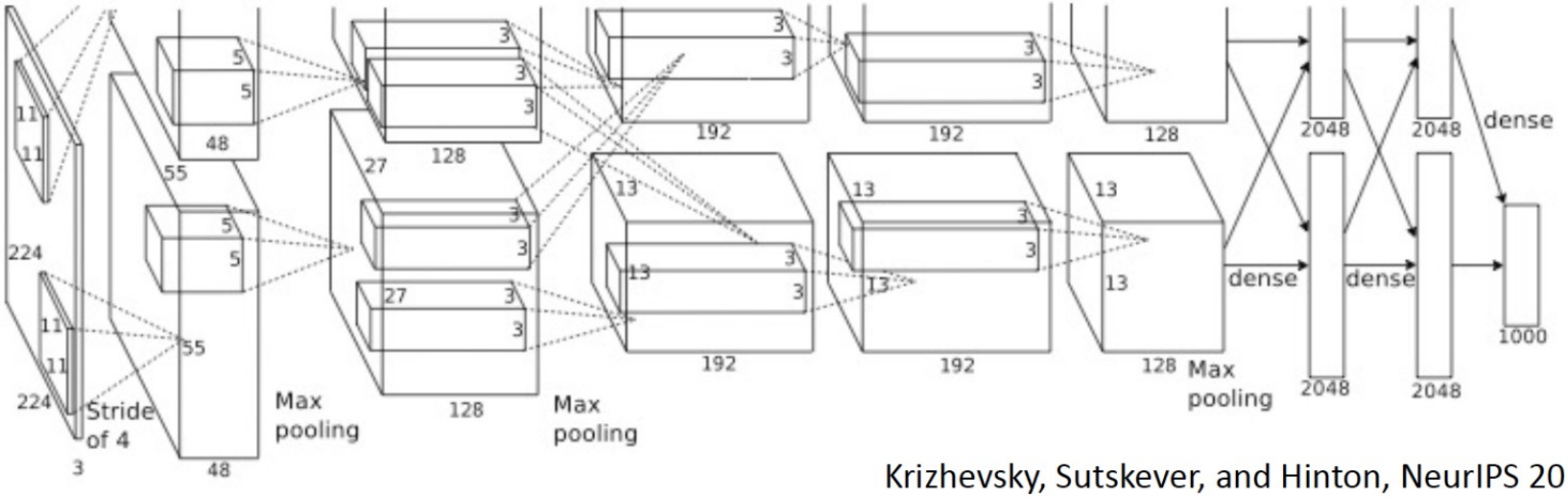
Very large dataset in computer vision which has been a benchmark. Images were labeled using **crowdsourcing**. “Olympics” of computer vision.



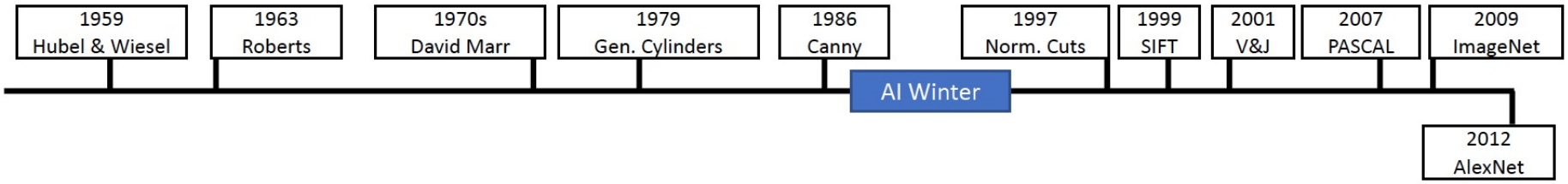


In 2012 a big advance happened at ILSVRC: error rate dropped from 25.8% to 16.4%.
From then on the error dropped significantly each year. Breakthrough point for deep learning.

AlexNet: Deep Learning Goes Mainstream



Krizhevsky, Sutskever, and Hinton, NeurIPS 2012



Seminal paper on deep learning which showed much better performance.

Perceptron

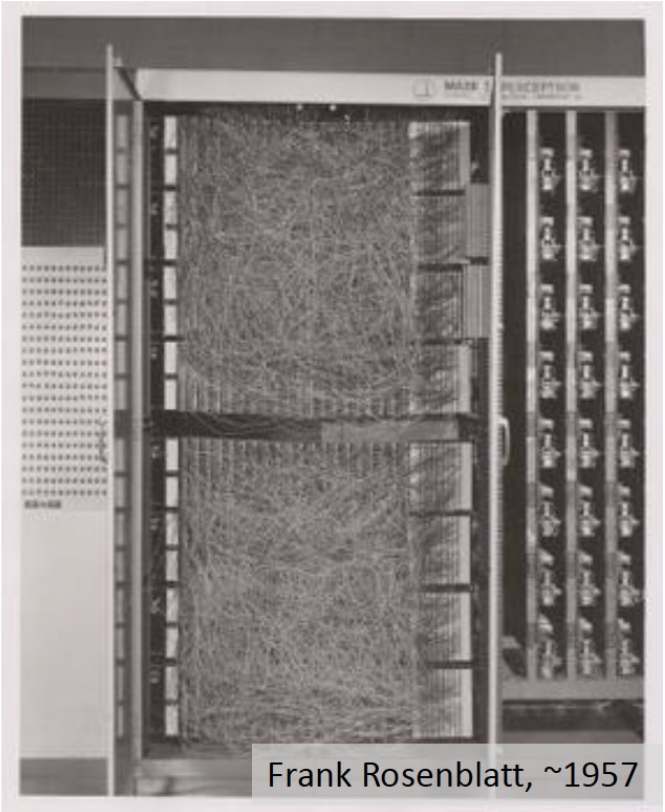
One of the earliest algorithms that could learn from data

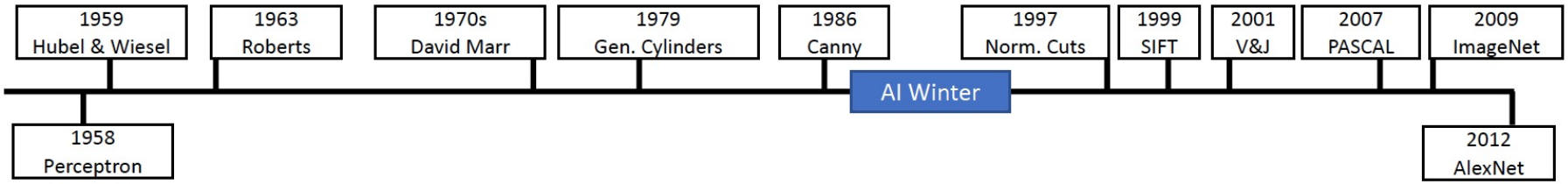
Implemented in hardware! Weights stored in potentiometers, updated with electric motors during learning

Connected to a camera that used 20x20 cadmium sulfide photocells to make a 400-pixel image

Could learn to recognize letters of the alphabet

Today we would recognize it as a **linear classifier**

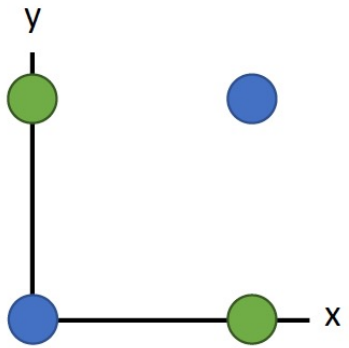

Frank Rosenblatt, ~1957

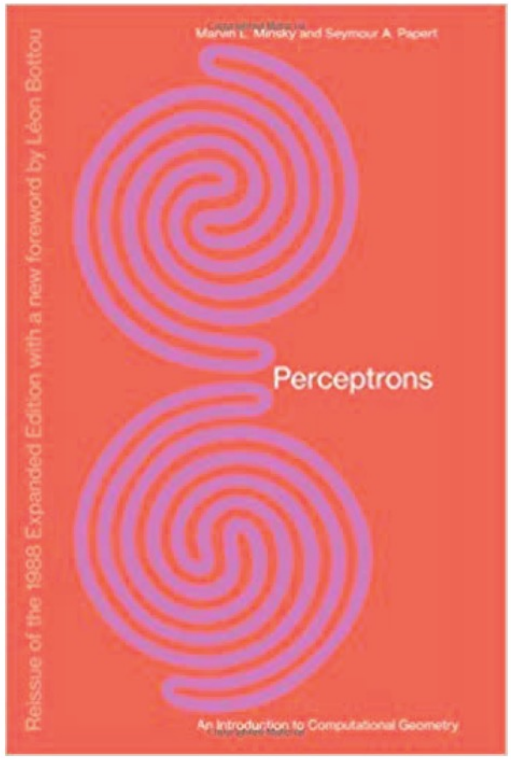


Perceptron: Cabinet size device. Weights were represented by potentiometers which were changed by motors during learning. Programming a computer was difficult.

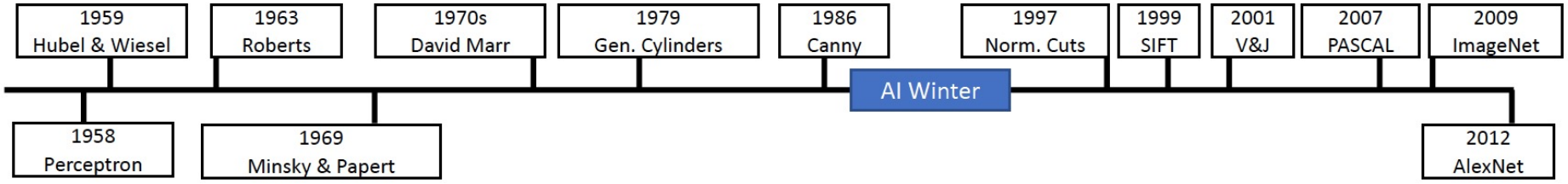
Minsky and Papert, 1969

X	Y	F(x,y)
0	0	0
0	1	1
1	0	1
1	1	0





Showed that Perceptrons could not learn the XOR function
Caused a lot of disillusionment in the field



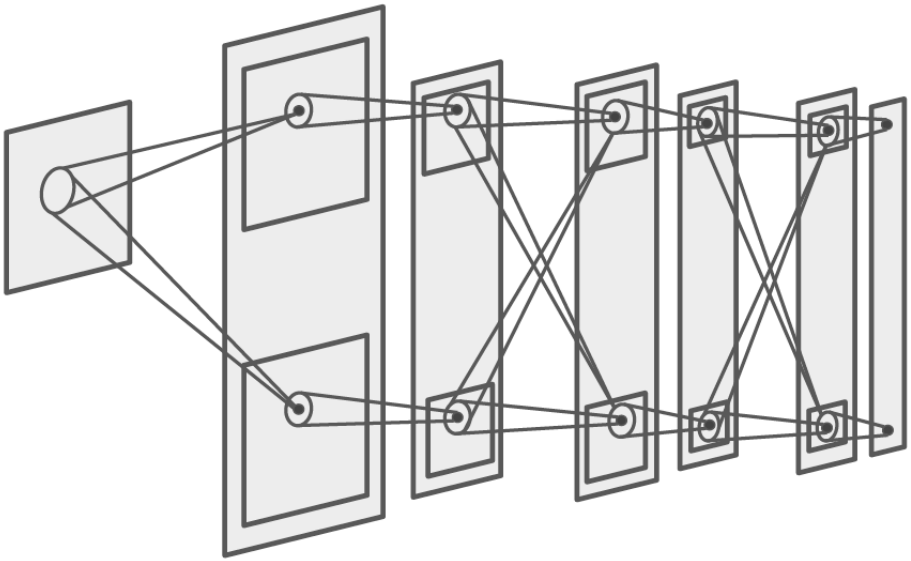
Since **perceptrons** had some limitations and researchers lost interest in them.
Multi-layer perceptron was mentioned to learn more complex tasks but did not attract attention

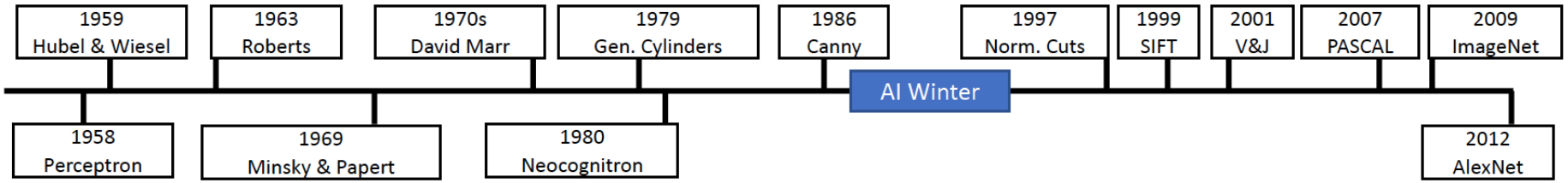
Neocognitron: Fukushima, 1980

Computational model the visual system, directly inspired by Hubel and Wiesel's hierarchy of complex and simple cells

Interleaved simple cells (convolution) and complex cells (pooling)

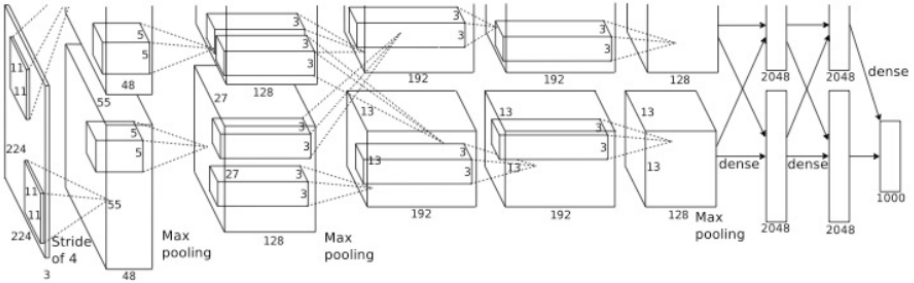
No practical training algorithm

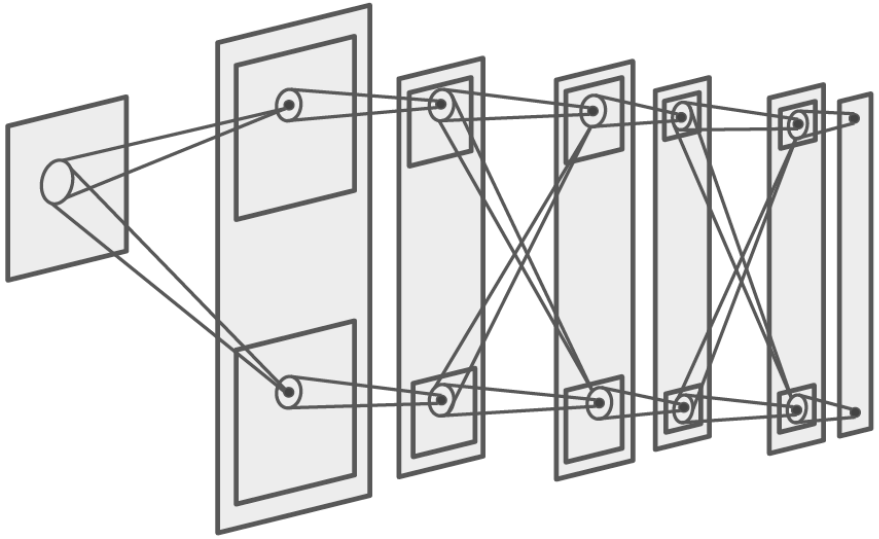




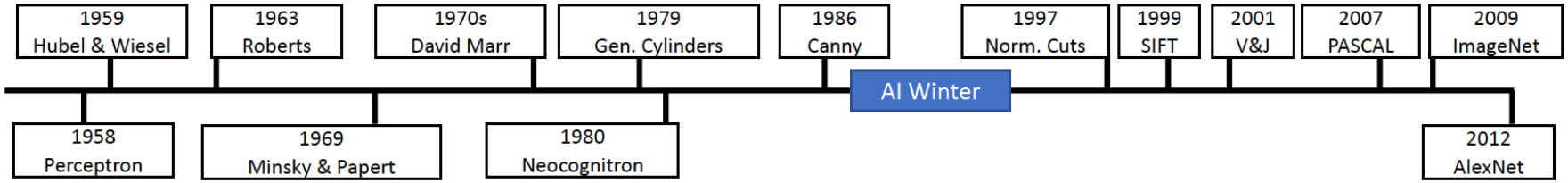
Disadvantage: no training algorithm to learn the parameters of the model.

Neocognitron: Fukushima, 1980





Looks a lot like AlexNet
more than 32 years later!



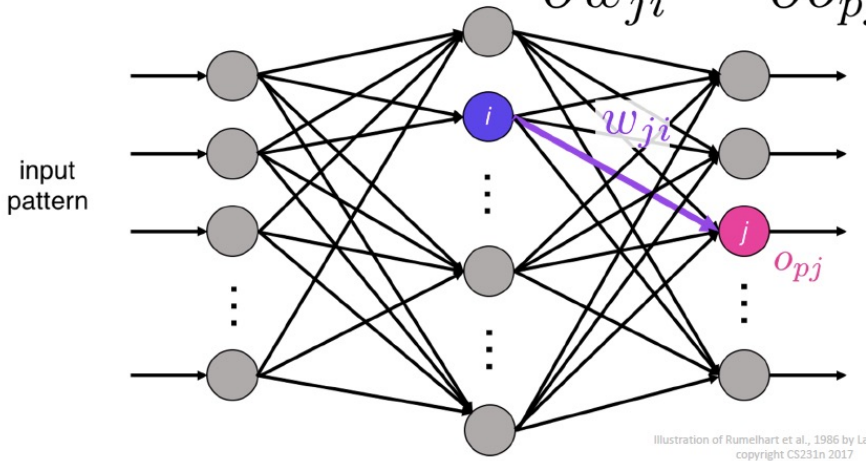
Backprop: Rumelhart, Hinton, and Williams, 1986

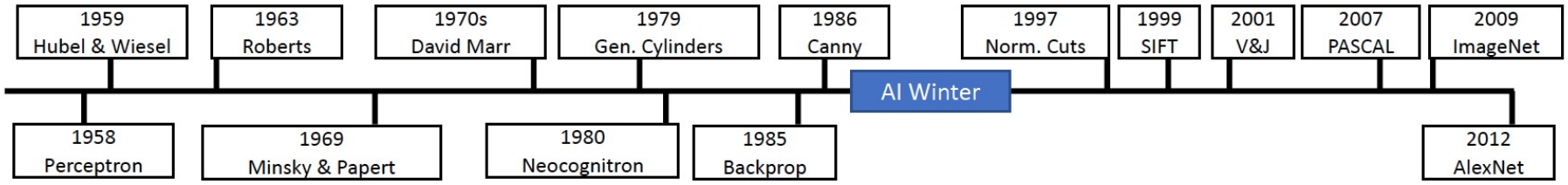
Introduced backpropagation for computing gradients in neural networks

Successfully trained perceptrons with multiple layers

$$\frac{\partial E_p}{\partial w_{ji}} = \frac{\partial E_p}{\partial o_{pj}} \frac{\partial o_{pj}}{\partial w_{ji}}$$

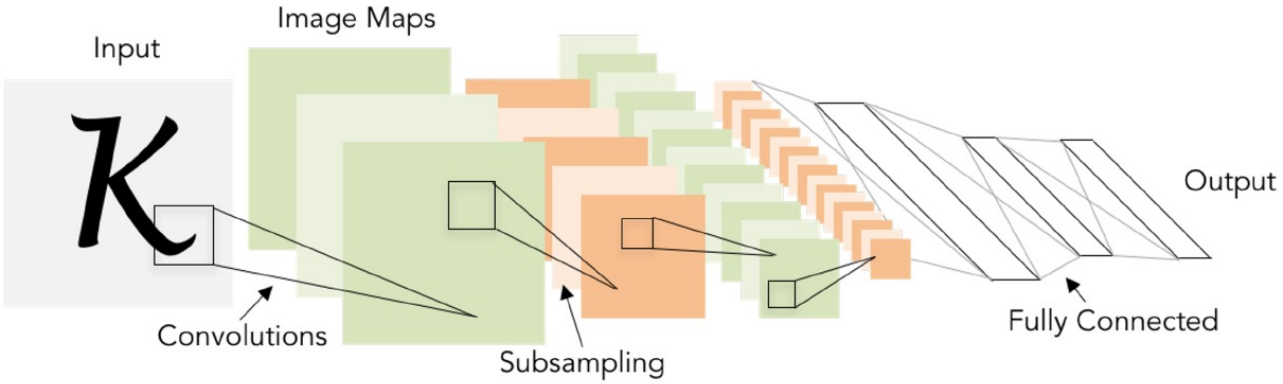
recognizable math



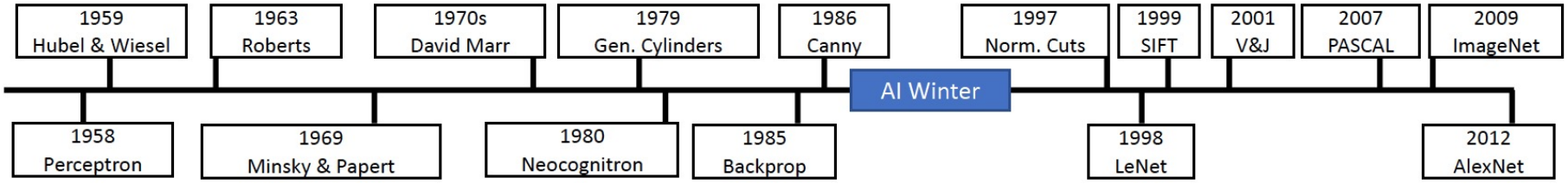


Proposed an algorithm for training multi-layer perceptrons.
Raised interest in neural networks again.

Convolutional Networks: LeCun et al, 1998



Applied backprop algorithm to a Neocognitron-like architecture
Learned to recognize handwritten digits
Was deployed in a commercial system by NEC, processed handwritten checks
Very similar to our modern convolutional networks!



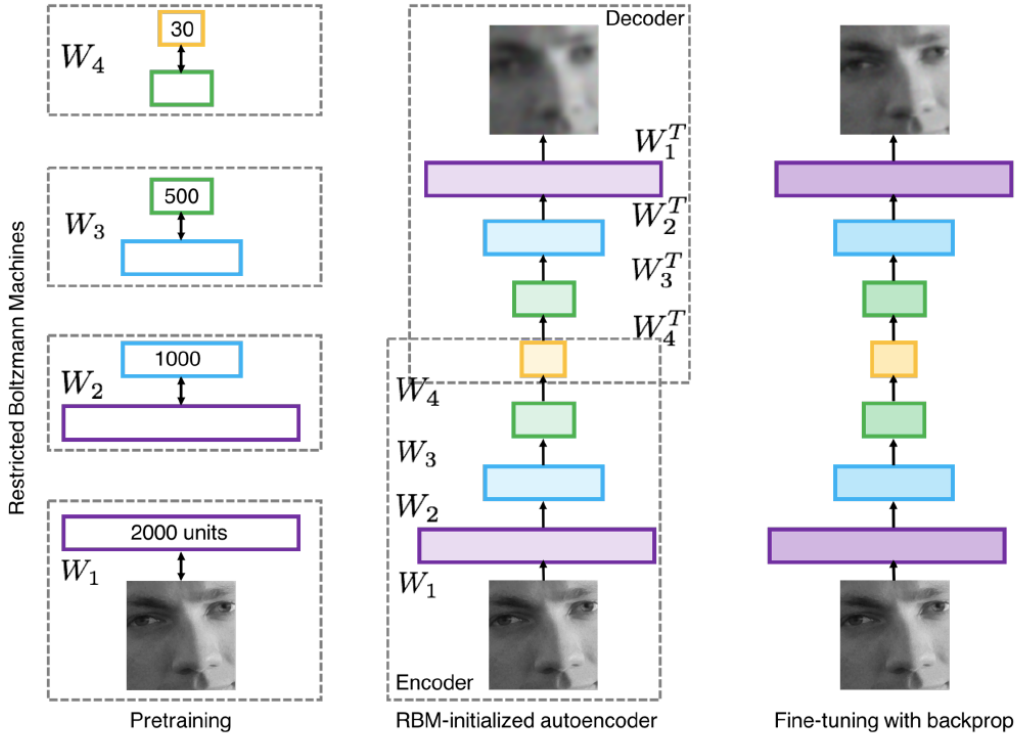
Combined Neocognitron method (1980) with backpropagation (1985) --> LeNet CNN arhitecture.

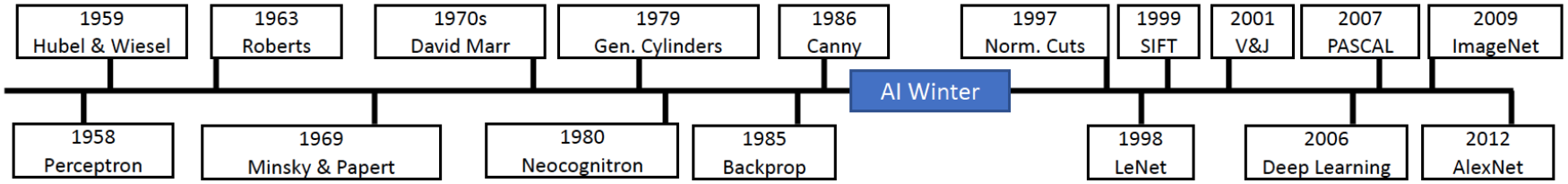
2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

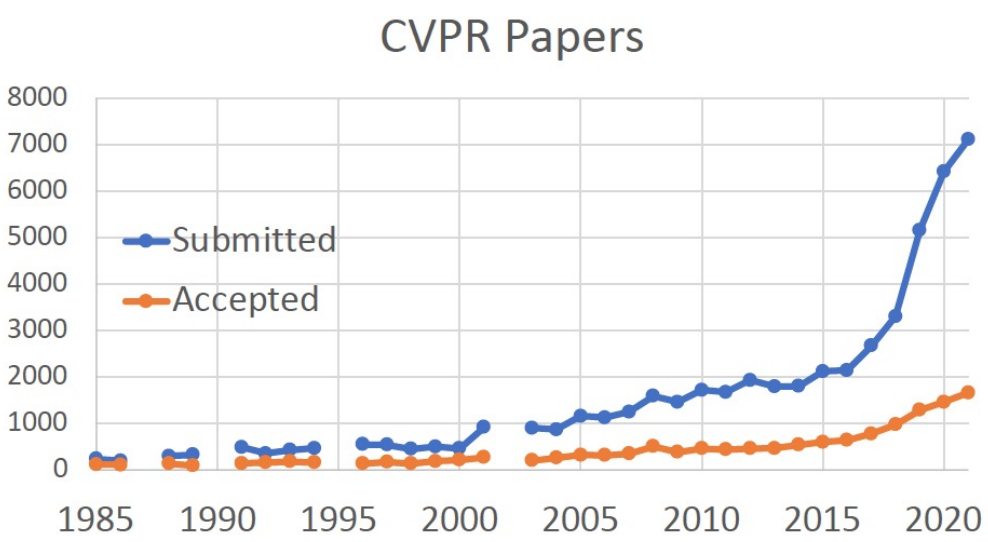
Not a mainstream research topic at this time

Hinton and Salakhutdinov, 2006
Bengio et al, 2007
Lee et al, 2009
Glorot and Bengio, 2010

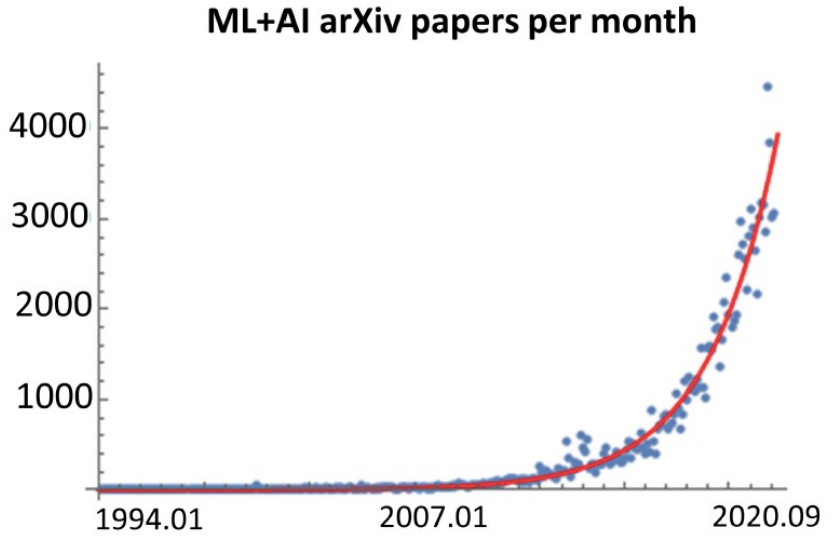




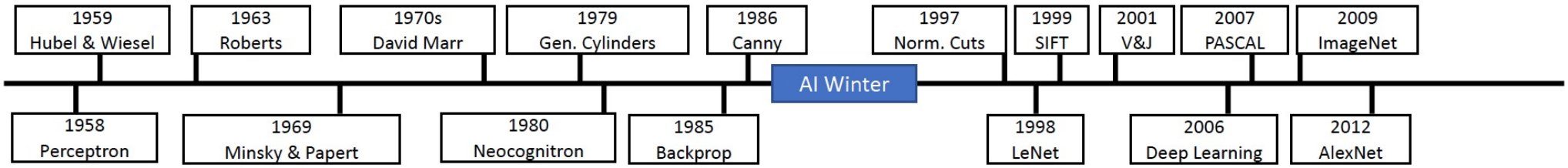
2012 to Present: Deep Learning Explosion



Publications at top Computer Vision conference



arXiv papers per month (source)

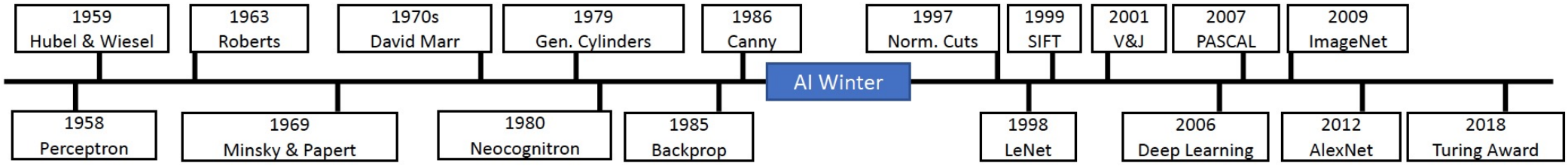


2018 Turing Award

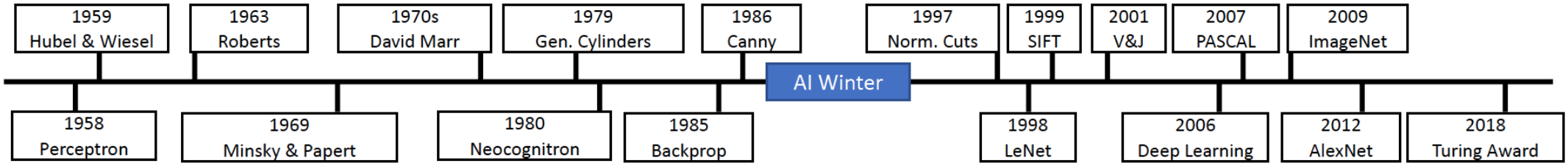

Yoshua Bengio


Geoffrey Hinton


Yann LeCun



"Nobel prize" of computer vision



ANNs are not new: summers and winters

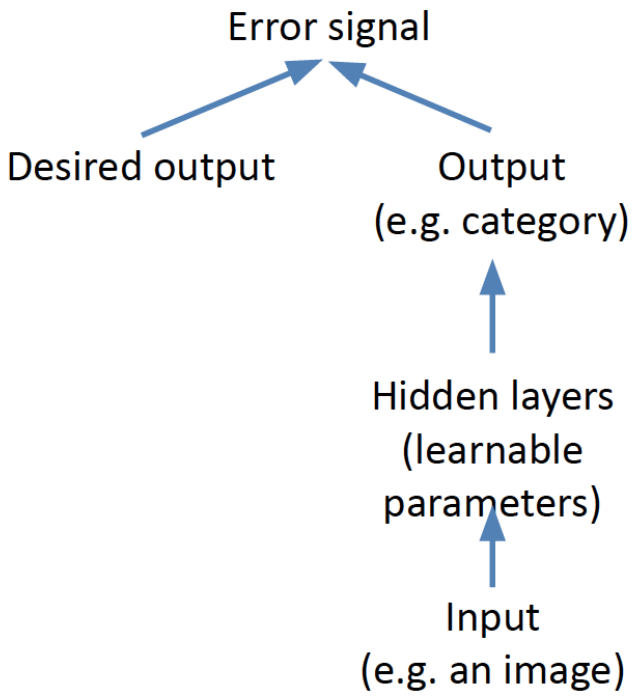
- F. Rosenblat, “The Perceptron: A Probabilistic Model For Information Storage and Organization in the Brain", Psychological Review, 1958.
- M. Minsky, S. Papert, “Perceptrons. An Introduction to Computational Geometry”, The MIT Press, 1969.
- With the advent of “backpropagation”, ANNs were popular again in late 80s, early 90s. [Rumelhart et al., 1986]
- Until the support vector machines and the kernel trick took over in mid 90s.
- With “deep learning,” ANNs are again back on stage from around 2010.
- Then, what was wrong in the 90s? And, what changed?





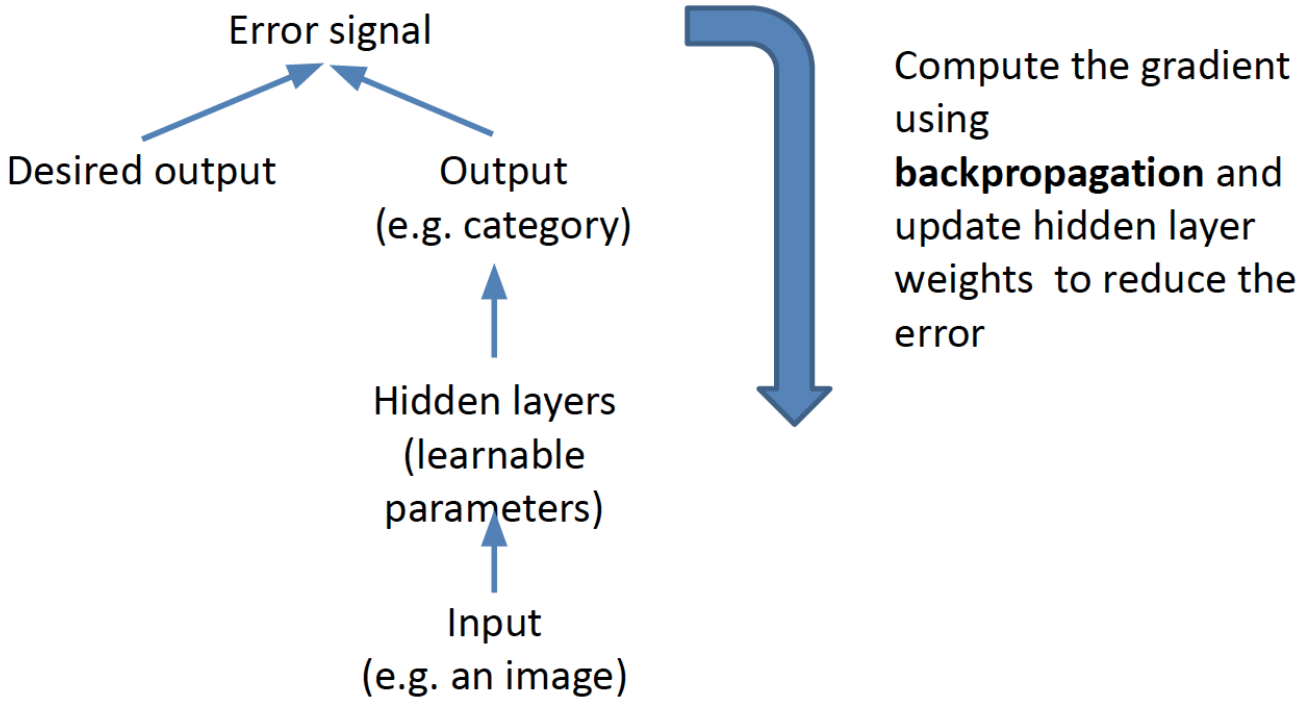


Backpropagation



Nothing but an application of the “chain rule”

Backpropagation



Nothing but an application of the “chain rule”

Why did people abandon backpropagation?

"People in machine learning had largely given up on it because:

- It did not seem to be able to make good use of multiple hidden layers
- It did not work well in recurrent networks"

Source: G. Hinton's talk at the Royal Society, May 22, 2015.
https://youtu.be/izrG86jycck

What was **actually** wrong with backpropagation?

"We all drew the wrong conclusions about why it failed. The real reasons were:

1. Our datasets were thousands of times too small.
2. Our computers were millions of times too slow.
3. We initialized the weights in a stupid way.
4. We used the wrong type of non-linearity."

Source: G. Hinton's talk at the Royal Society, May 22, 2015.
https://youtu.be/izrG86jycck

DL: the comeback of neural networks

In 2009, G. Hinton and his students developed a method to train a deep network for speech recognition.

First, they trained each layer (one layer at a time) in an unsupervised manner.

Then, they added a final layer of supervision and then ran backpropagation.

DL: the comeback of neural networks

This deep network outperformed a long standing state-of-the-art on a medium sized speech recognition dataset.

Mohamed, A. R., Dahl, G. E. and Hinton, G. E. “Deep belief networks for phone recognition.” NIPS workshop on deep learning for speech recognition, 2009.

DL: the comeback of neural networks

This speech recognition deep network was in use in the Android operating system as early as 2012.

The second success story

In computer vision.

ILSVRC 2012: A competition.

1.2 million images, 1000 categories

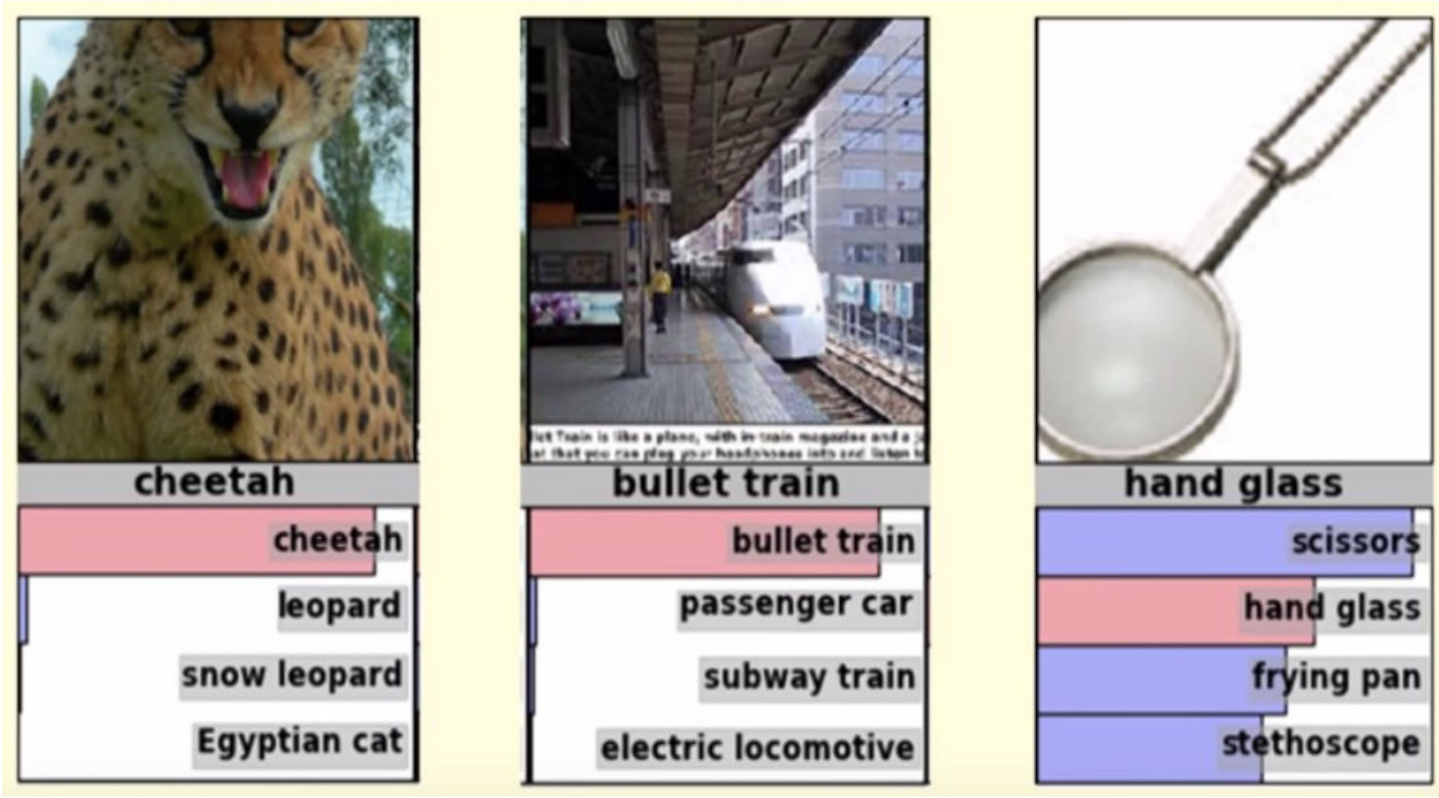
The task: given an image, make 5 predictions for the dominant object in the image. If one of them is correct, then it is counted as a success.

The second success story

G. Hinton's student Alex Krizhevsky trained a 7-layer convolutional neural network (which is now known as **AlexNet**) on 2 GPUs. [Krizhevsky, Sutskever & Hinton 2012]

Used the same trick (Mohamed, Dahl, Hinton 2009) to train the network.

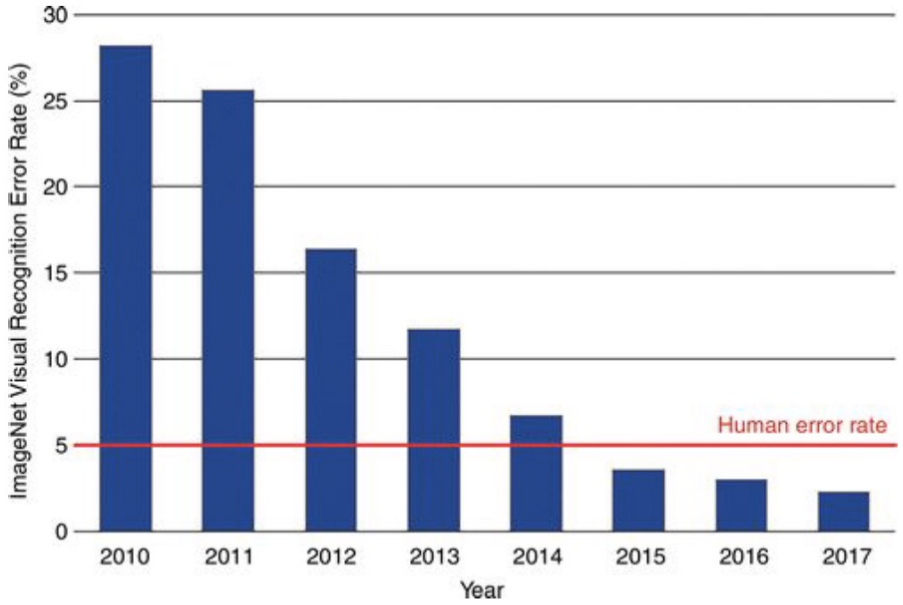
The second success story



Source: G. Hinton's talk at the Royal Society, May 22, 2015.
https://youtu.be/izrG86jycck

The second success story

Since then (2012), deep learning based participants dominated the ILSVRC competitions.



[Figure from Langlotz et al. 2019]

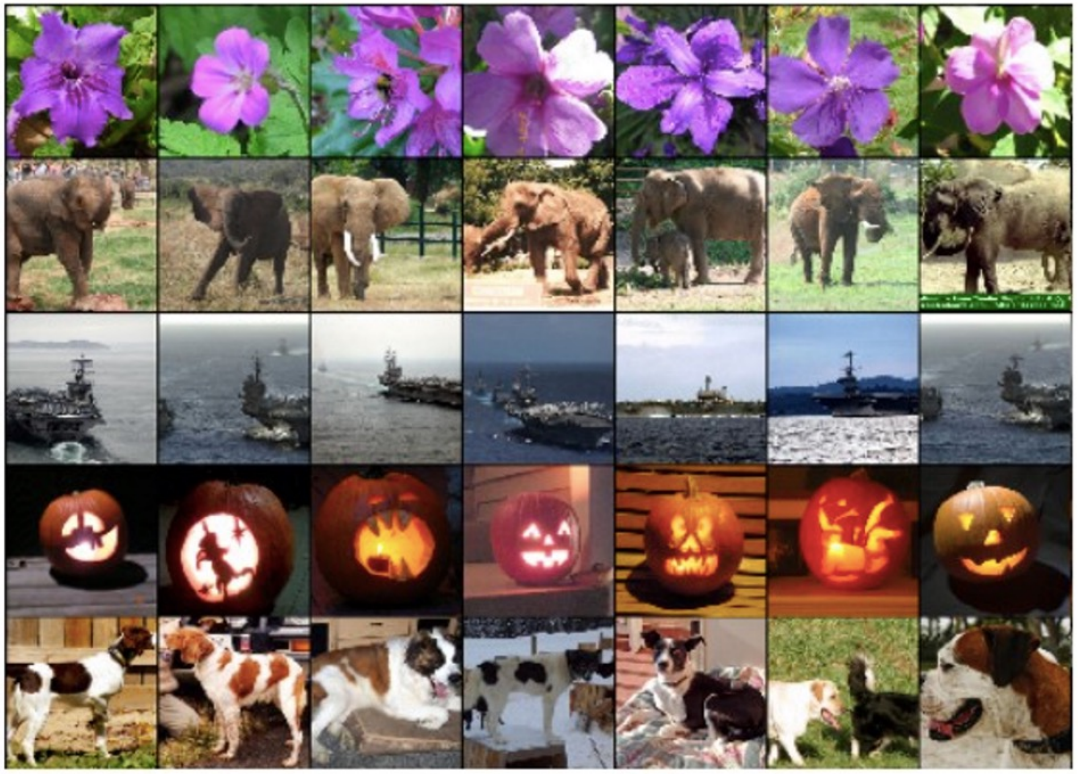
And, many other success stories so far.

2012 to Present: Deep Learning is Everywhere

Image Classification



Image Retrieval



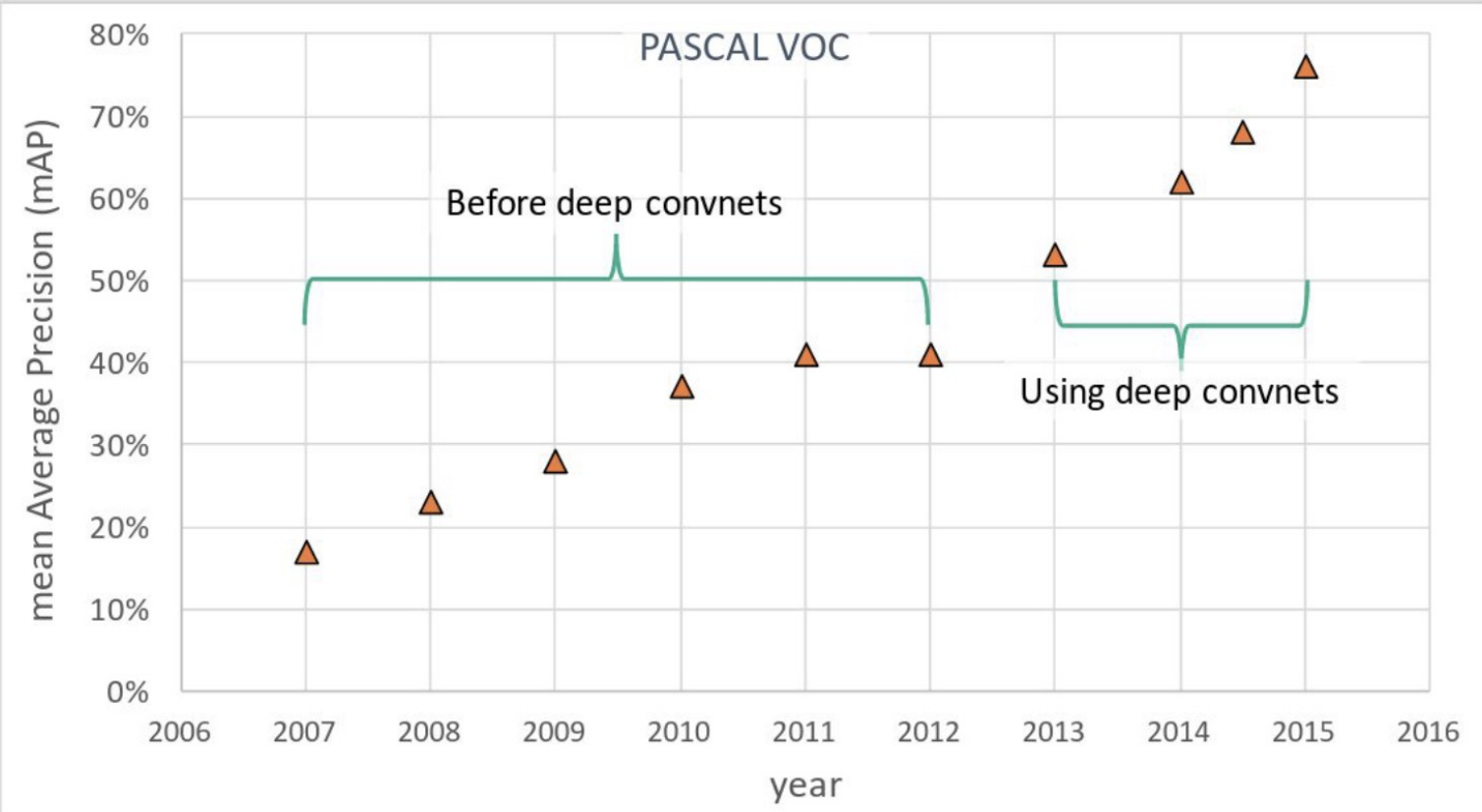
Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

Object detection



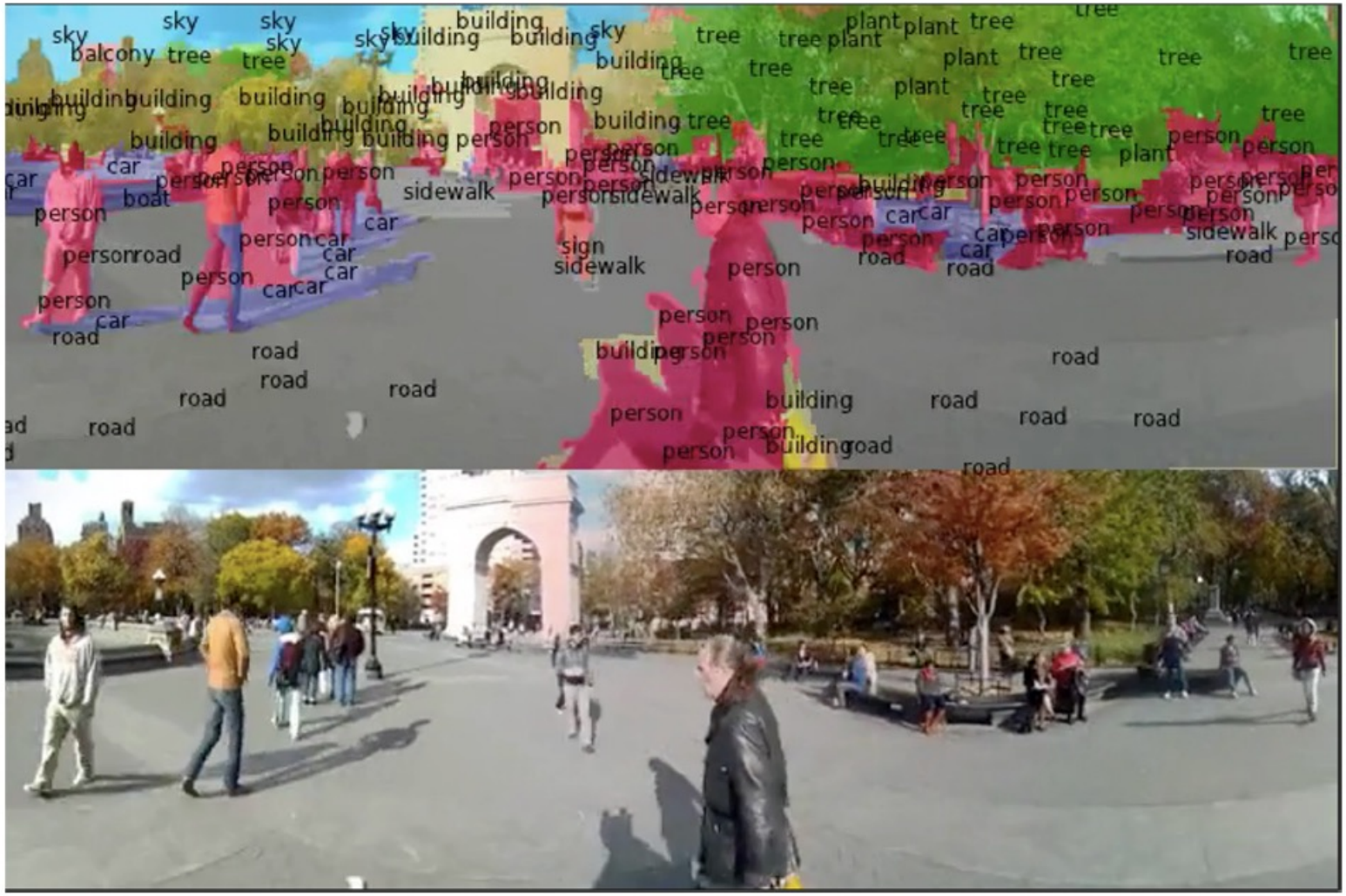
(Figure from Ren, He, Girschik, Sun, NIPS 2015)

Object detection



(Figure from “Fast RCNN slides” by R. Girschik)

Image Segmentation



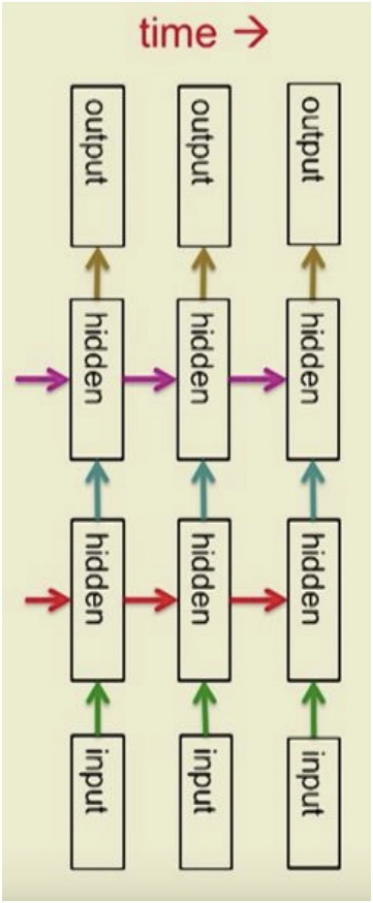
Fabaret et al, 2012

Machine Translation

State-of-the-art machine translation using Recurrent Neural Networks (RNNs)

Using a combination of Encoder and Decoder RNNs

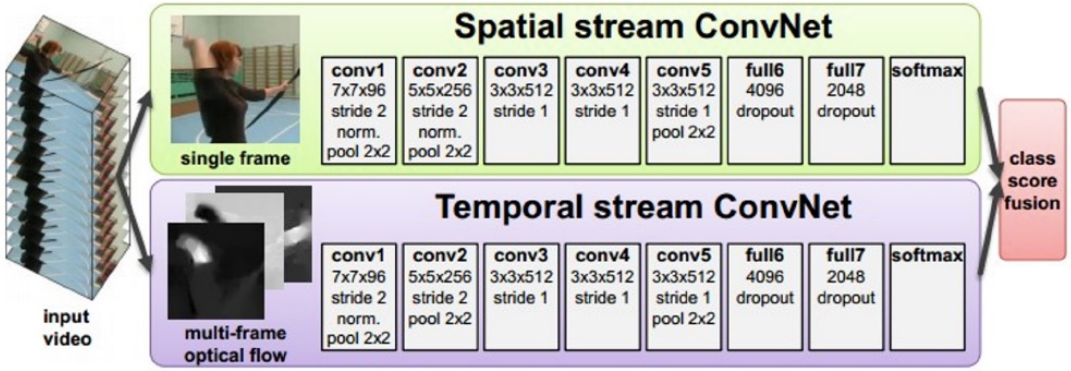
[Sutskever, Vinyals and Le 2014]



(Figure from Hinton's Royal Society talk)

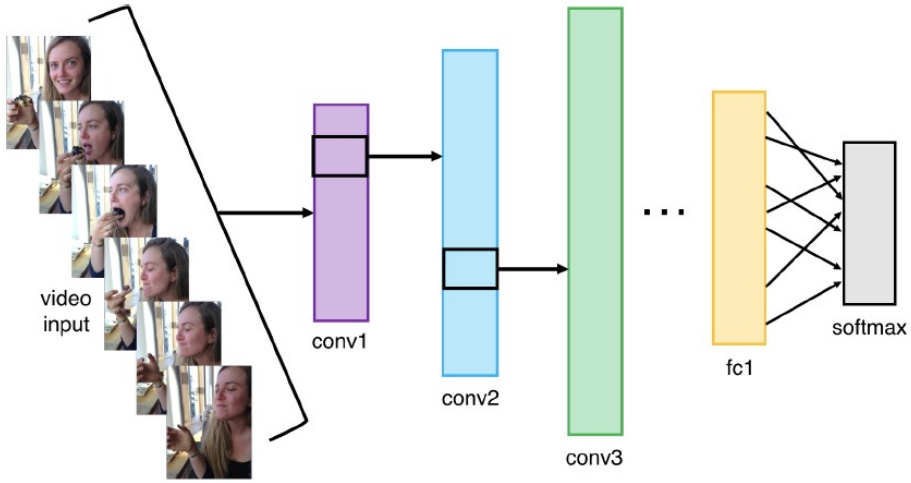
2012 to Present: Deep Learning is Everywhere

Video Classification



Simonyan et al, 2014

Activity Recognition

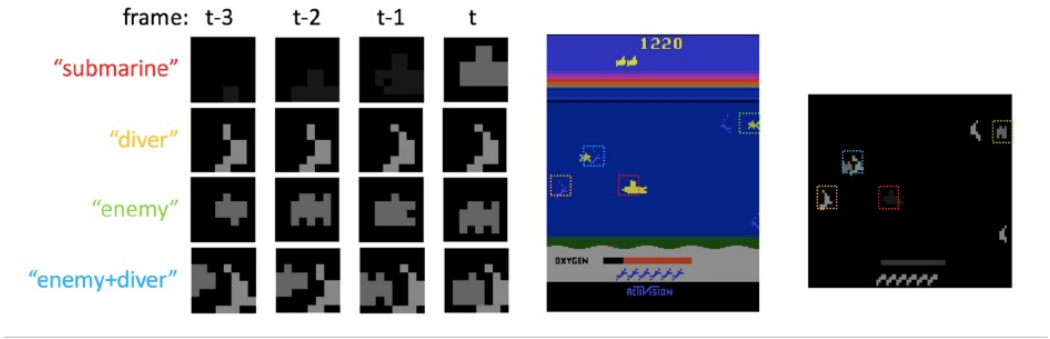


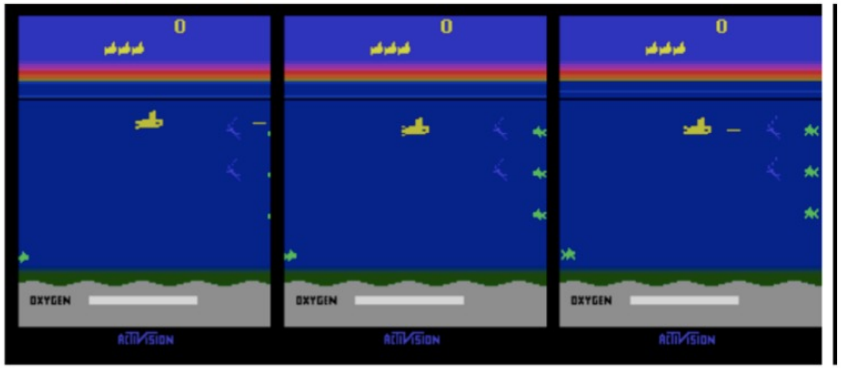
2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshev and Szegedy, 2014)



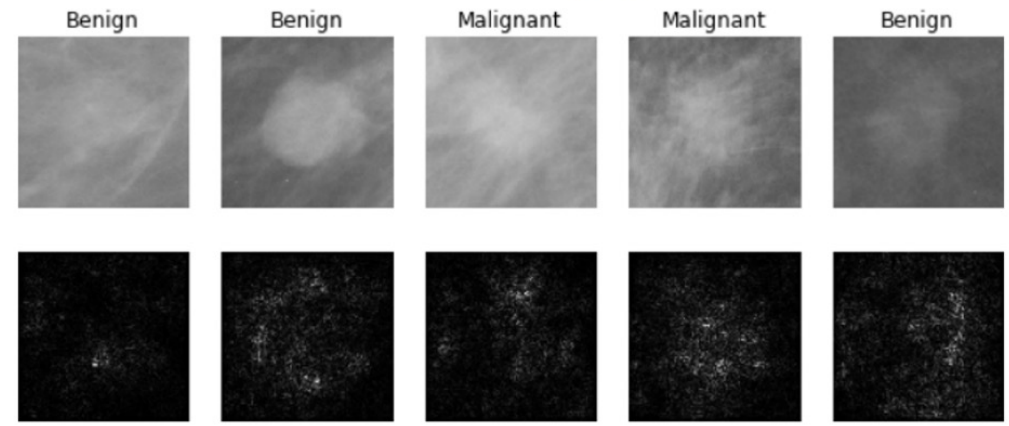
Playing Atari games (Guo et al, 2014)





2012 to Present: Deep Learning is Everywhere

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification



Dieleman et al, 2014

From left to right: public domain by NASA, usage permitted by ESA/Hubble, public domain by NASA, and public domain.

Whale recognition



Kaggle Challenge

This image by Christin Khan is in the public domain and originally came from the U.S. NOAA.

CNNs are also used for tasks at which humans are not good ad: e.g. medical diagnosis

2012 to Present: Deep Learning is Everywhere



A white teddy bear sitting in the grass



A man in a baseball uniform throwing a ball



A woman is holding a cat in her hand

Image Captioning

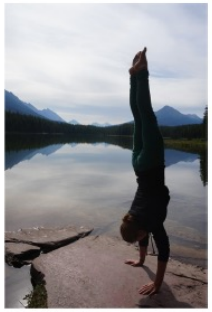
Vinyals et al, 2015
Karpathy and Fei-Fei, 2015



A man riding a wave on top of a surfboard



A cat sitting on a suitcase on the floor



A woman standing on a beach holding a surfboard

All images are CC0 Public domain:
https://pixabay.com/en/luggage-antique-cat-1643010/
https://pixabay.com/en/teddy-plush-bears-cute-teddy-bear-1623436/
https://pixabay.com/en/surf-wave-summer-sport-litoral-1668716/
https://pixabay.com/en/woman-female-model-portrait-adult-983967/
https://pixabay.com/en/handstand-lake-meditation-496008/
https://pixabay.com/en/baseball-player-shortstop-infield-1045263/

Captions generated by Justin Johnson using Neuraltalk2

Image captioning: writing natural language descriptions about images.

Image captioning



A group of people shopping at an outdoor market.

(GT: People are crouched around in an open market.)

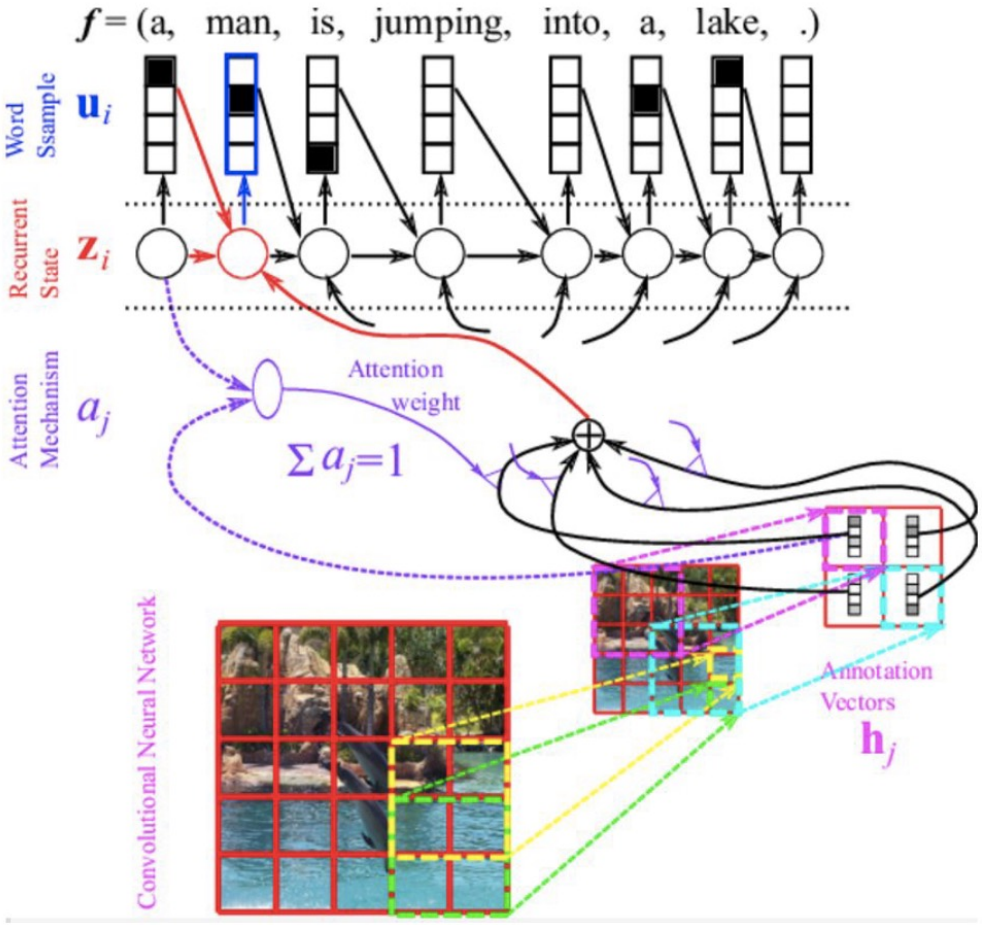


A close up of a child holding a stuffed animal

(GT: A young girl asleep on the sofa cuddling a stuffed bear.)

(Figure from Hinton's Royal Society talk)

Image captioning



[Donahue et al. 2015]

Generative deep networks

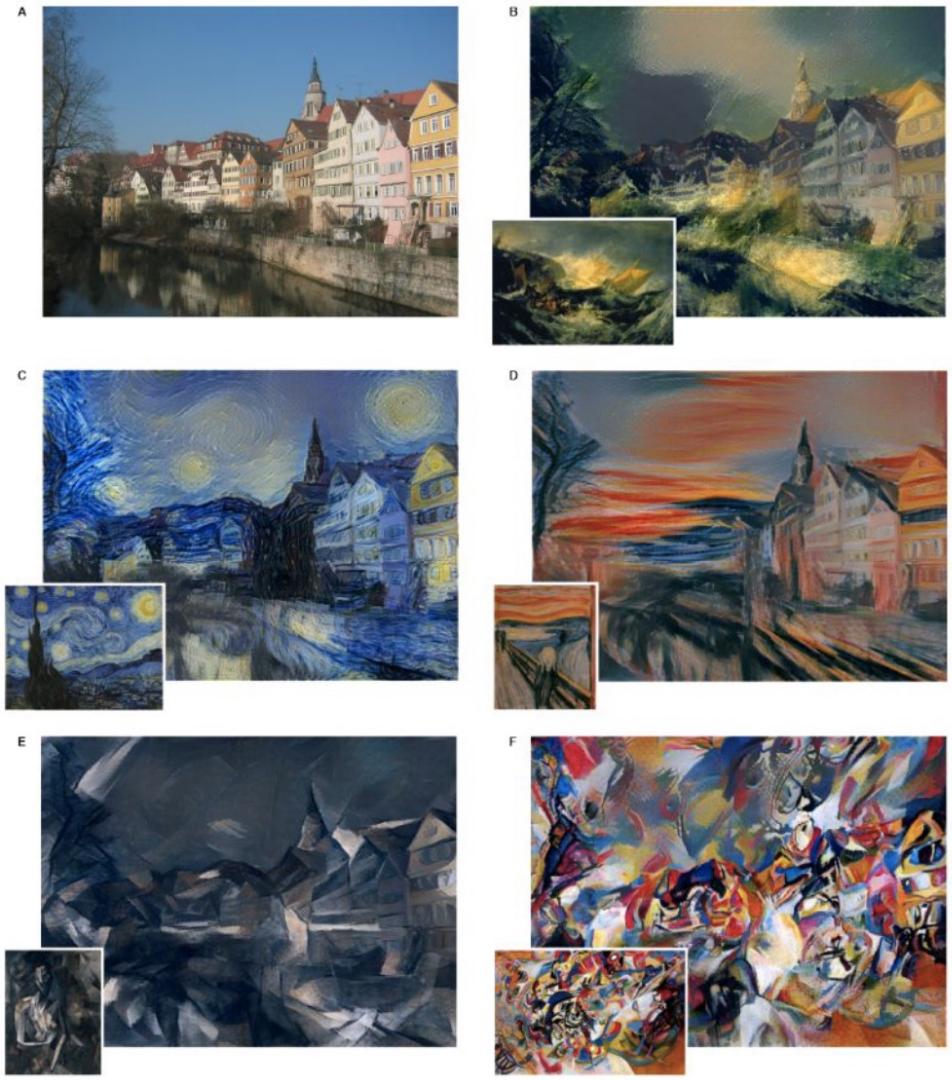
Image generation:



Figure 1. Image completions sampled from a PixelRNN.

(From Oord, Kalchbrenner, Kavukcuoglu 2016)

Style transfer



(Figure from Gatys, Ecker, Bethge 2015)

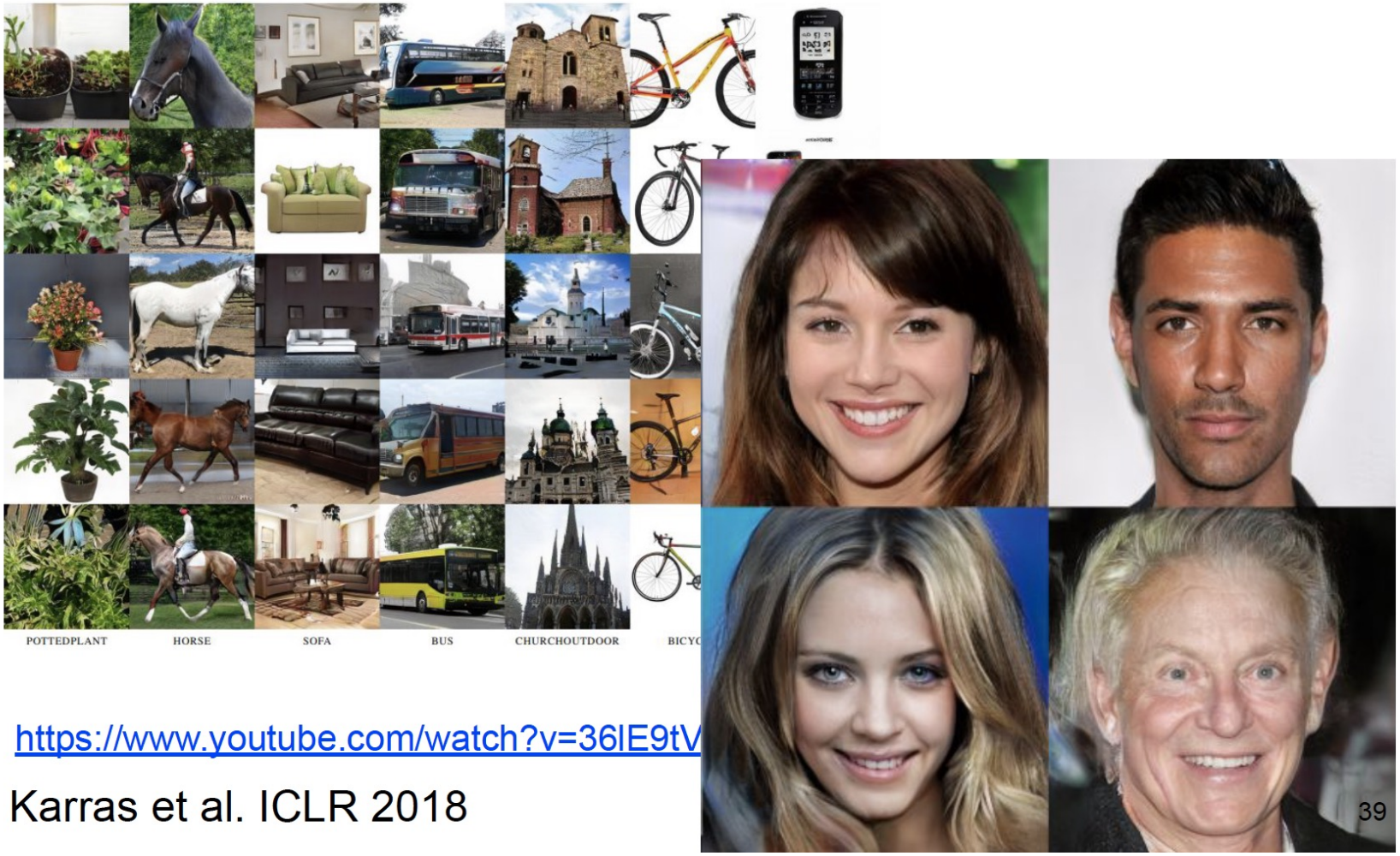
Style Transfer





Painted in the style of Van Gogh's Starry Night.
Generated by Deep Dream (http://deepdreamgenerator.com/)

Generating images



https://www.youtube.com/watch?v=36lE9tV

Karras et al. ICLR 2018

2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of an avocado. an armchair imitating an avocado.

AI-GENERATED IMAGES



amesh et al, “DALL·E: Creating Images from Text”, 2021. https://openai.com/blog/dall-e/

2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of a peach. an armchair imitating a peach.

AI-GENERATED IMAGES



Ramesh et al, "DALL·E: Creating Images from Text", 2021. https://openai.com/blog/dall-e/

https://openai.com/blog/dall-e/

… and videos



https://makeavideo.studio/

[Singer et al. 2022]

https://makeavideo.studio/

Deep learning beats doctors at spotting eye diseases. [De Fauw et al. Nature Medicine 2018]

Deep Learning for Electronic Health Records
[Rajkomar et al. NPJ Digital Medicine 2018]

End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography [Ardila et al. Nature Medicine 2018]

Self-driving cars



[Figure from
https://www.wikiwand.com/en/Self-driving_car]

Computer vision and natural language processing has been taken over by DL.

And Many More…

- Music generation:

https://www.youtube.com/watch?&v=0VTI1BBLydE

- Large language models (LLMs), e.g. BERT, GPT, ChatGPT

https://openai.com/blog/chatgpt/

Applications of LLMs

- Copywriting
- Code generation and auto-complete
- Shell command generation
- Regex generation
- Automated code reviews & code quality improvement
- Database query optimization
- DevOps assistance / automation
- Frontend / website generation
- Personalized recommendations
- Chatbot / support agent assist
- Grammar correction & style
- Translation
- …

[From https://leighmariebraswell.substack.com/p/overview-and-applications-of-large]

- What enabled deep learning to become popular?


Algorithms
224
5
11
11
55
27
3
3
13
13
3
3
192
13
13
3
3
192
13
13
128
Max
dense
dense
1000
2048
2048
Data
Computation
GTX

Why deep learning? Why now?

- Because the field is guided by experimental findings rather than by theory, algorithmic advances only become possible when appropriate data and hardware are available to try new ideas (or to scale up old ideas, as is often the case).
- Machine learning isn't mathematics or physics, where major advances can be done with a pen and a piece of paper. It's an engineering science.
- The **real bottlenecks throughout the 1990s and 2000s were data and hardware**.
- But here's what happened during that time: the internet took off and high-performance graphics chips were developed for the needs of the gaming market.

Hardware

- Between 1990 and 2010, off-the-shelf CPUs became faster by a factor of approximately 5,000.
- As a result, nowadays it's possible to run small deep learning models on your laptop, whereas this would have been intractable 25 years ago.
- But typical deep learning models used in computer vision or speech recognition require orders of magnitude more computational power than your laptop can deliver.
- Throughout the 2000s, companies like NVIDIA and AMD invested billions of dollars in developing fast, massively parallel chips (graphical processing units, or GPUs) to power the graphics of increasingly photorealistic video games—cheap, single-purpose supercomputers designed to render complex 3D scenes on your screen in real time.
- This investment came to benefit the scientific community when, in 2007, NVIDIA launched CUDA (https://developer.nvidia.com/about-cuda), a programming interface for its line of GPUs.

Hardware

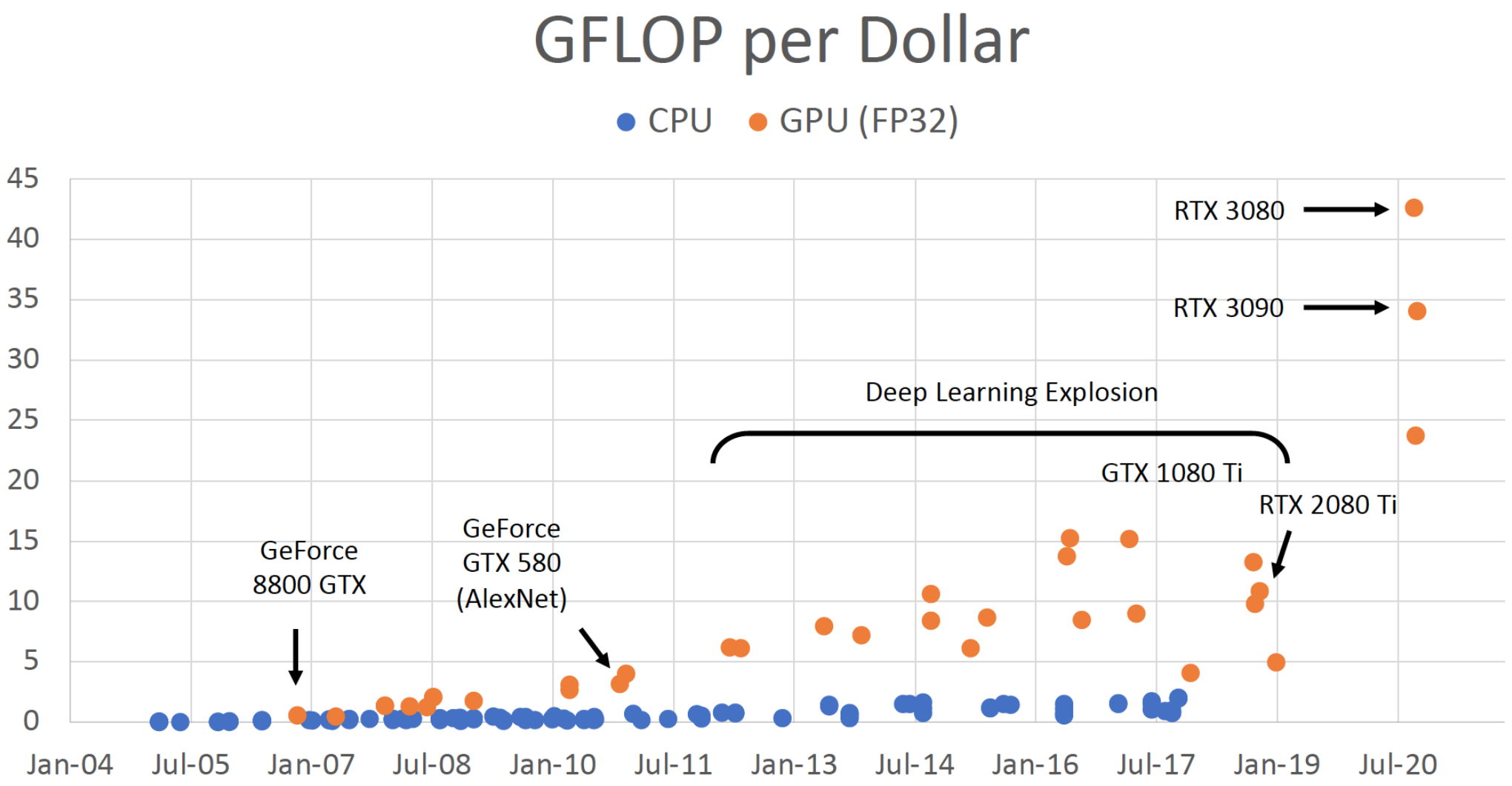
- A small number of GPUs started replacing massive clusters of CPUs in various highly parallelizable applications, beginning with physics modeling.
- **Deep neural networks**, consisting mostly of many small **matrix multiplications**, are **also highly parallelizable**, and around 2011 some researchers began to write CUDA implementations of neural nets—Dan Ciresan and Alex Krizhevsky were among the first.
- What happened is that the gaming market subsidized supercomputing for the next generation of artificial intelligence applications.
- Sometimes, big things begin as games.
- Today, the NVIDIA Titan RTX, a GPU that cost $2,500 at the end of 2019, can deliver a peak of 16 teraFLOPS in single precision (16 trillion `float32` operations per second).
- That's about **500 times more computing power than the world's fastest supercomputer from 1990, the Intel Touchstone Delta**.
- On a Titan RTX, it takes only a few hours to train an ImageNet model of the sort that would have won the ILSVRC competition around 2012 or 2013.
- Meanwhile, large companies train deep learning models on clusters of hundreds of GPUs.

Hardware

- What's more, the deep learning industry has been moving beyond GPUs and is investing in increasingly specialized, efficient chips for deep learning.
- In 2016, at its annual I/O convention, Google revealed its Tensor Processing Unit (**TPU**) project: a new chip design developed from the ground up to run deep neural networks significantly faster and far more energy efficient than top-of-the-line GPUs.
- Today, in 2020, the third iteration of the TPU card represents 420 teraFLOPS of computing power.
- That's 10,000 times more than the Intel Touchstone Delta from 1990.
- These TPU cards are designed to be assembled into large-scale configurations, called "pods."
- One pod (1024 TPU cards) peaks at 100 petaFLOPS. For scale, that's about 10% of the peak computing power of the current largest supercomputer, the IBM Summit at Oak Ridge National Lab, which consists of 27,000 NVIDIA GPUs and peaks at around 1.1 exaFLOPS.

mega-	M	10^{6}
giga-	G	10^{9}
tera-	T	10^{12}
peta-	P	10^{15}
exa-	E	10^{18}
zetta-	Z	10^{21}
yotta-	Y	10^{24}

https://www.techtarget.com/searchstorage/definition/Kilo-mega-giga-tera-peta-and-all-that#:~:text=The%20prefixes%20exa%20(1%20quintillion,Units%20(SI)%20in%201975.



GPU computing power has increased exponentially since 2012.

Data

- AI is sometimes heralded as **the new industrial revolution**.
- If **deep learning is the steam engine** of this revolution, then **data is its coal**: the raw material that powers our intelligent machines, without which nothing would be possible.
- In addition to the exponential progress in storage hardware over the past 20 years, the **game changer has been the rise of the internet**, making it **feasible to collect and distribute very large datasets** for machine learning.
- Today, large companies work with image datasets, video datasets, and natural language datasets that couldn't have been collected without the internet.
  - User-generated image tags on Flickr, for instance, have been a treasure trove of data for computer vision.
  - So are YouTube videos.
  - Wikipedia is a key dataset for natural language processing.

Algorithms

- In addition to hardware and data, until the late 2000s, we were missing a reliable way to train very deep neural networks.
- As a result, neural networks were still fairly shallow, using only one or two layers of representations; thus, they weren't able to shine against more-refined shallow methods such as SVMs and random forests.
- The key issue was that of gradient propagation through deep stacks of layers.
- The **feedback signal used to train neural networks would fade away as the number of layers increased**.

Algorithms

- This changed around 2009–2010 with the advent of several simple but important algorithmic improvements that allowed for better gradient propagation:
  - **Better activation functions** for neural layers
  - **Better weight-initialization schemes**,
  - **Better optimization schemes**, such as RMSProp and Adam
- Only when these improvements began to allow for training models with 10 or more layers did deep learning start to shine.
- Finally, in 2014, 2015, and 2016, even more advanced ways to improve gradient propagation were discovered, such as **batch normalization**, **residual connections**, and depthwise separable convolutions.

Algorithms

- Today, we can train models that are arbitrarily deep from scratch.
- This has unlocked the use of extremely large models, which hold considerable representational power—that is to say, which encode very rich hypothesis spaces.
- This extreme scalability is one of the defining characteristics of modern deep learning.
- Large-scale model architectures, which feature tens of layers and tens of millions of parameters, have brought about critical advances both in computer vision (for instance, architectures such as ResNet, Inception, or Xception) and natural language processing (for instance, large Transformer-based architectures such as BERT, GPT-3, or XLNet).

A new wave of investment

- As deep learning became the new state of the art for computer vision in 2012–2013, and eventually for all perceptual tasks, industry leaders took note.
- What followed was a gradual wave of industry investment far beyond anything previously seen in the history of AI.

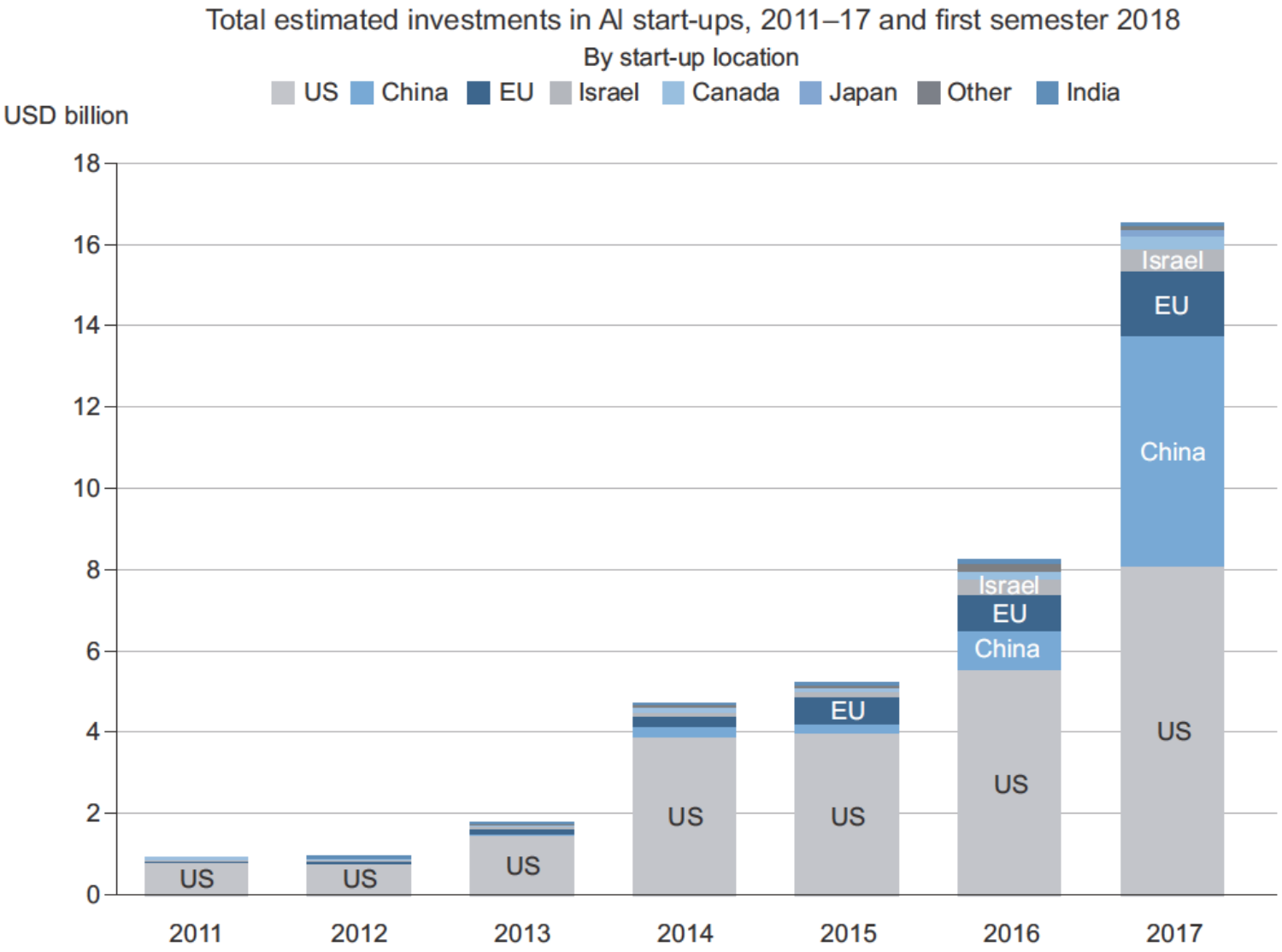


Figure 1.14 OECD estimate of total investments in AI startups (Source: http://mng.bz/zGN6)

The democratization of deep learning

- One of the key factors driving this inflow of new faces in deep learning has been the democratization of the toolsets used in the field. In the early days, doing deep learning required significant C++ and CUDA expertise, which few people possessed.
- Nowadays, basic Python scripting skills suffice to do advanced deep learning research.
- This has been driven most notably by the development of the TensorFlow library—a symbolic tensor-manipulation framework for Python that support autodifferentiation, greatly simplifying the implementation of new models—and by the rise of user-friendly libraries such as Keras, which makes deep learning as easy as manipulating LEGO bricks.
- After its release in early 2015, Keras quickly became the go-to deep learning solution for large numbers of new startups, graduate students, and researchers pivoting into the field.

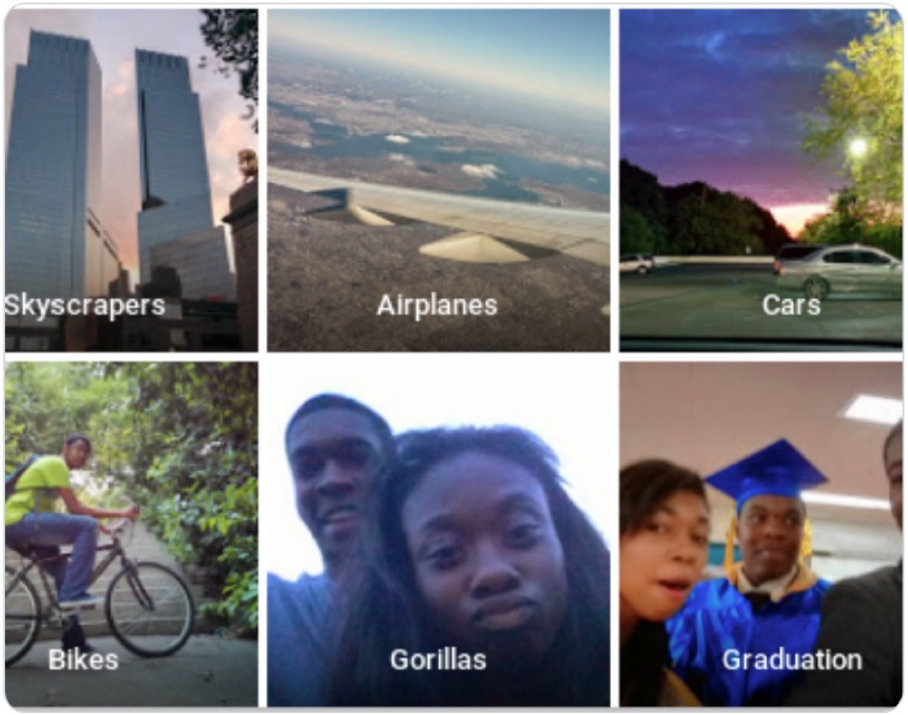
Will it last?

- Will we still be using deep neural networks in 20 years?
- We may not be using neural networks two decades from now, but whatever we use will directly inherit from modern deep learning and its core concepts.
- These important properties can be broadly sorted into three categories:
  - **Simplicity**—Deep learning removes the need for feature engineering, replacing complex, brittle, engineering-heavy pipelines with simple, end-to-end trainable models that are typically built using only five or six different tensor operations.
  - **Scalability**—Deep learning is highly amenable to parallelization on GPUs or TPUs. In addition, deep learning models are trained by iterating over small batches of data, allowing them to be trained on datasets of arbitrary size.
  - **Versatility and reusability**—Unlike many prior machine learning approaches, deep learning models can be trained on additional data without restarting from scratch, making them viable for continuous online learning—an important property for very large production models. Furthermore, trained deep learning models are repurposable and thus reusable: for instance, it's possible to take a deep learning model trained for image classification and drop it into a video processing pipeline.

Despite our success, computer vision still has a long way to go…

Computer Vision can cause harm

Harmful Stereotypes



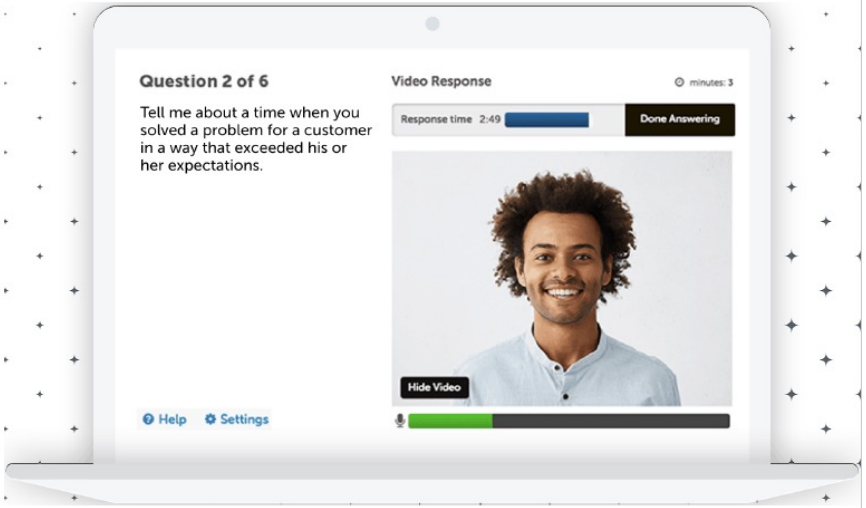
Barocas et al, "The Problem With Bias: Allocative Versus Representational Harms in Machine Learning", SIGCIS 2017
Kate Crawford, "The Trouble with Bias", NeurIPS 2017 Keynote
Source: https://twitter.com/jackyalcine/status/615329515909156865 (2015)

Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Source: https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/
https://www.hirevue.com/platform/online-video-interviewing-software
Example Credit: Timnit Gebru

Generative AI on Trial

- A group of artists filed a class-action lawsuit in a United States court against Stability AI and Midjourney, companies that provide image generators, and DeviantArt, an online community that hosts its own image generator. The lawsuit claims that the models' ability to generate work "in the style of" a given artist infringes artists' intellectual property rights and harms them financially.
- In a separate action, writer, programmer, and lawyer Matthew Butterick brought a class-action claim against Microsoft, OpenAI, and GitHub in a U.S. court. The plaintiff alleges that Copilot, a model that generates computer code, outputs open-source code without properly crediting its creators.
- Getty Images announced its intent to sue Stability AI in a British court for using images scraped from Getty's collection to train its models.
- **We're thinking:** Laws that protect intellectual property haven't yet caught up with AI. Without legal clarity, engineers have less freedom to innovate, and investors have less certainty about which approaches to support.

Source: The Batch Newsletter by https://www.deeplearning.ai/



This image is copyright-free United States government work

Example credit:
Andrej Karpathy


Computer Vision Technology
Can Better Our Lives

The modern machine learning landscape

- A great way to get a sense of the current landscape of machine learning algorithms and tools is to look at machine learning competitions on **Kaggle** (http://kaggle.com).
- Kaggle offers a realistic way to assess what works and what doesn't:
  - Highly competitive environment (some contests have thousands of entrants and million dollar prizes)
  - Wide variety of machine learning problems covered
  - What kind of algorithm is reliably winning competitions?
  - What tools do top entrants use?

Kaggle

- In early 2019, Kaggle ran a survey asking teams that ended in the top five of any competition since 2017 which primary software tool they had used in the competition.
- It turns out that top teams tend to use either deep learning methods (most often via the Keras library) or gradient boosted trees (most often via the LightGBM or XGBoost libraries).

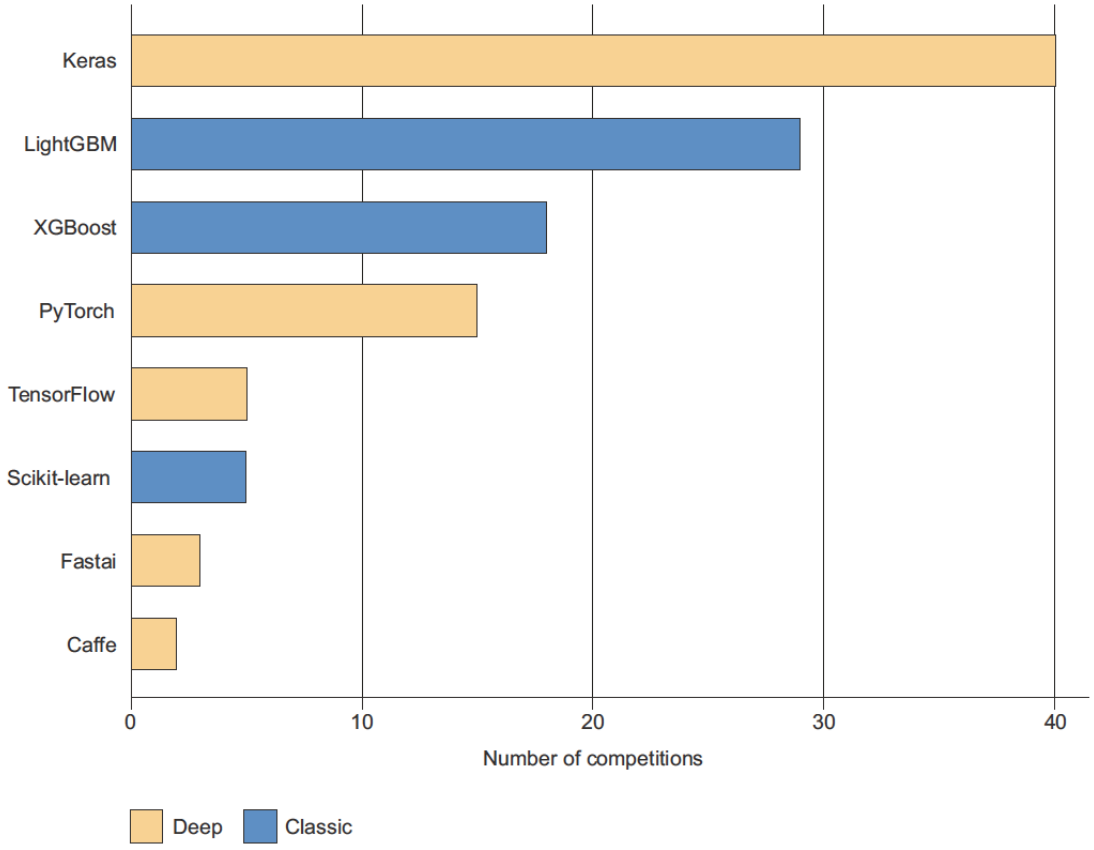


Figure 1.12 Machine learning tools used by top teams on Kaggle

Worldwide

- Kaggle also runs a yearly survey among machine learning and data science professionals worldwide.
- With tens of thousands of respondents, this survey is one of our most reliable sources about the state of the industry.
- Figure shows the percentage of usage of different machine learning software frameworks.

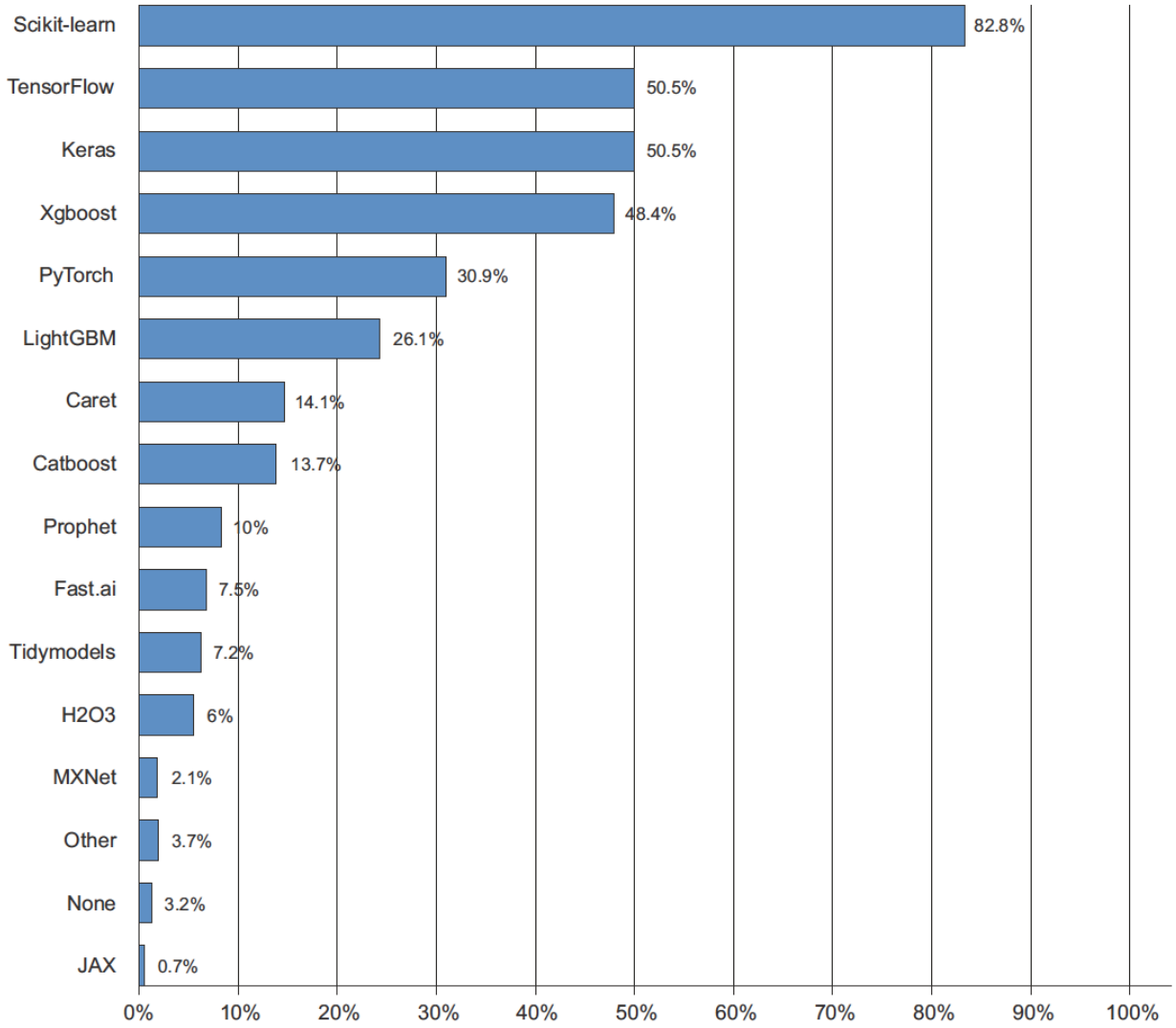


Figure 1.13 Tool usage across the machine learning and data science industry (Source: www.kaggle.com/kaggle-survey-2020)

Kaggle

- From 2016 to 2020, the entire machine learning and data science industry has been dominated by these two approaches: **deep learning** and **gradient boosted trees**.
- Specifically, **gradient boosted trees** is used for problems where **structured data** is available (shallow learning problems), whereas **deep learning** is used for **perceptual problems such as image classification**.
- Users of gradient boosted trees tend to use **Scikit-learn, XGBoost**, or **LightGBM**.
- Meanwhile, most practitioners of deep learning use **Keras**, often in combination with its parent framework TensorFlow.
- The common point of these tools is they're all Python libraries: Python is by far the most widely used language for machine learning and data science.

References

- Deep Learning with Python, F. Chollet, 202, Chapter 1
- Lecture notes of Dr. Justin Johnson (University of Michigan)
  - EECS 498/598 Deep Learning for Computer Vision
  - https://web.eecs.umich.edu/~justincj/teaching/eecs498/WI2022/
  - https://www.youtube.com/watch?v=dJYGatp4SvA&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r
- Lecture notes of Dr. Emre Akbaş (METU)
  - Ceng 501 – Deep Learning
  - https://user.ceng.metu.edu.tr/~emre/Fall2022-DeepLearning.html

1. Ardila, D., Kiraly, A.P., Bharadwaj, S., Choi, B., Reicher, J.J., Peng, L., Tse, D., Etemadi, M., Ye, W., Corrado, G. and Naidich, D.P., 2019. End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography. Nature medicine, 25(6), p.954.
2. Cybenko, G. (1989). Approximation by superpositions of a sigmoidal function. Mathematics of control, signals and systems, 2(4), 303-314.
3. De Fauw, J., Ledsam, J.R., Romera-Paredes, B., Nikolov, S., Tomasev, N., Blackwell, S., Askham, H., Glorot, X., O'Donoghue, B., Visentin, D. and van den Driessche, G., 2018. Clinically applicable deep learning for diagnosis and referral in retinal disease. Nature Medicine, 24(9), p.1342.
4. Donahue, J., Anne Hendricks, L., Guadarrama, S., Rohrbach, M., Venugopalan, S., Saenko, K., & Darrell, T. (2015). Long-term recurrent convolutional networks for visual recognition and description. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 2625-2634).
5. Mitchel, Tom. (1997). Machine Learning, McGraw Hill.
6. Mohamed, A. R., Dahl, G., & Hinton, G. (2009, December). Deep belief networks for phone recognition. In Nips workshop on deep learning for speech recognition and related applications (Vol. 1, No. 9, p. 39).
7. Gatys, L. A., Ecker, A. S., & Bethge, M. (2015). A neural algorithm of artistic style. arXiv preprint arXiv:1508.06576.
8. Girschik, R. (2015). Slides for Fast RCNN. https://www.dropbox.com/s/vlyrkgd8nz8gy5l/fast-rcnn.pdf?dl=0

8. Hinton, G. (2015). Talk at the Royal Society, May 22, 2015. https://youtu.be/izrG86jycck
9. Karras, T., Aila, T., Laine, S. and Lehtinen, J., 2017. Progressive growing of gans for improved quality, stability, and variation. ICLR 2018.
10. Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In Advances in neural information processing systems (pp. 1097-1105).
11. Kurach, K., Andrychowicz, M., & Sutskever, I. (2015). Neural random-access machines. arXiv preprint arXiv:1511.06392.
12. Langlotz, C. P., Allen, B., Erickson, B. J., Kalpathy-Cramer, J., Bigelow, K., Cook, T. S., ... & Kandarpa, K. (2019). A roadmap for foundational research on artificial intelligence in medical imaging: from the 2018 NIH/RSNA/ACR/The Academy Workshop. Radiology, 291(3), 781.
13. Nguyen, A., Yosinski, J., & Clune, J. (2015, June). Deep neural networks are easily fooled: High confidence predictions for unrecognizable images. In 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 427-436). IEEE.
14. Oord, A. V. D., Kalchbrenner, N., & Kavukcuoglu, K. (2016). Pixel recurrent neural networks. arXiv preprint arXiv:1601.06759.
15. Rajkomar, A., Oren, E., Chen, K., Dai, A.M., Hajaj, N., Hardt, M., Liu, P.J., Liu, X., Marcus, J., Sun, M. and Sundberg, P., 2018. Scalable and accurate deep learning with electronic health records. NPJ Digital Medicine, 1(1), p.18.
16. Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster R-CNN: Towards real-time object detection with region proposal networks. In Advances in neural information processing systems (pp. 91-99).
17. Rosenblatt, Frank (1957). "The Perceptron—a perceiving and recognizing automaton". Report 85-460-1. Cornell Aeronautical Laboratory.
18. Rumelhart, D E.; Hinton, G.; Williams, R. (1986a). "Learning representations by back-propagating errors". Nature. 323 (6088): 533–536.
19. Singer, U., Polyak, A., Hayes, T., Yin, X., An, J., Zhang, S., ... & Taigman, Y. (2022). Make-A-Video: Text-to-Video Generation without Text-Video Data. arXiv preprint arXiv:2209.14792.
20. Sutskever, I., Vinyals, O., & Le, Q. V. (2014). Sequence to sequence learning with neural networks. In Advances in neural information processing systems (pp. 3104-3112).